

Fundamentals of the density functional theory (DFT의 기초)

이재동 (JAEDONG LEE)

Quantum Dynamics and Information Laboratory
Dept. of Physics and Chemistry, DGIST, Daegu, Korea

Biography

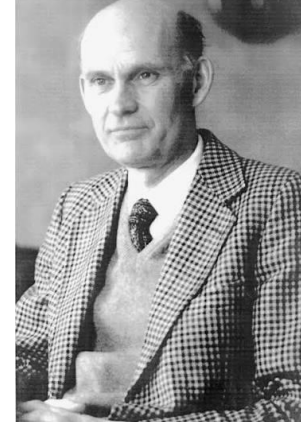
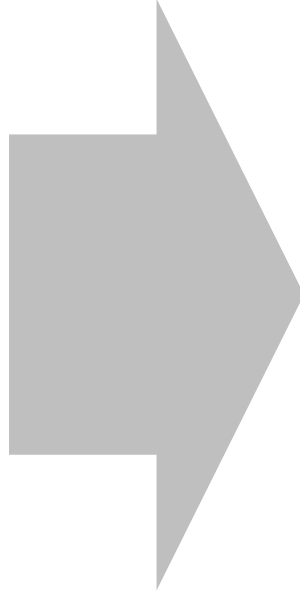


APW / FLAPW

Arthur Freeman



ByungIl Min



GWA

Lars Hedin



Olle Gunnarsson

0. Structure of matter
1. Many-body theory of electron system
2. Density functional theory
3. Exchange and correlation in DFT
4. Practical solving of electronic problem

0. Structure of matter

1. Many-body theory of electron system

2. Density functional theory

3. Exchange and correlation in DFT

4. Practical solving of electronic problem

0. Structure of matter

Matter - interacting electrons and nuclei (ions)

$$H = H_e + H_I + H_{eI}$$

$$H_e = T_e + V_{ee} = -\frac{\hbar^2}{2m} \sum_i \nabla_i^2 + \frac{e^2}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

$$H_I = T_I + V_{II} = -\frac{\hbar^2}{2M} \sum_I \nabla_I^2 + \frac{e^2}{2} \sum_{I \neq J} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|}$$

$$H_{eI} = -e^2 \sum_{i,I} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|}$$

$$H = H_e + H_n + H_{eI} + \underbrace{(\text{relativistic terms})}$$

mass velocity, Darwin, spin-orbit, ...

0. Structure of matter

Adiabatic approximation (Born-Oppenheimer approximation)

$$H\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N; \mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N; \mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_N) \equiv E\Psi(\mathbf{r}, \mathbf{R})$$

- many-body systems consisting of a number of electrons and nuclei
- Schrödinger equation is too difficult to solve

Born-Oppenheimer approximation says :

- electrons are about 1800 times lighter than nuclei
- motions of electrons are much faster than those of nuclei
- the approximation divides complicated motion into two parts
 - motion of nuclei
 - motion of electron

0. Structure of matter

Adiabatic approximation (Born-Oppenheimer approximation)

$$(T_e + V_{ee} + V_{eI}) \psi_n(\mathbf{r}, \mathbf{R}) = \varepsilon_n(\mathbf{R}) \psi_n(\mathbf{r}, \mathbf{R}) \quad ; \quad \mathbf{R} \text{ fixed}$$

$$\Psi(\mathbf{r}, \mathbf{R}) = \sum_n \phi_n(\mathbf{R}) \psi_n(\mathbf{r}, \mathbf{R})$$

$$H\Psi(\mathbf{r}, \mathbf{R}) = (T_e + V_{ee} + V_{eI} + T_I + V_{II}) \sum_n \phi_n(\mathbf{R}) \psi_n(\mathbf{r}, \mathbf{R}) = E \sum_n \phi_n(\mathbf{R}) \psi_n(\mathbf{r}, \mathbf{R})$$

$$(T_I + V_{II} + \varepsilon_n(\mathbf{R})) \phi_n(\mathbf{R}) \psi_n(\mathbf{r}, \mathbf{R}) = E \phi_n(\mathbf{R}) \psi_n(\mathbf{r}, \mathbf{R})$$

$$= \psi_n(\mathbf{r}, \mathbf{R}) (T_I + V_{II} + \varepsilon_n(\mathbf{R})) \phi_n(\mathbf{R}) + \left[-\frac{\hbar^2}{2M} \sum_I (\nabla_I^2 \psi_n(\mathbf{r}, \mathbf{R}) + 2 \nabla_I \psi_n(\mathbf{r}, \mathbf{R}) \nabla_I) \phi_n(\mathbf{R}) \right]$$

(nuclear dynamics does not cause electronic transition)

$$(T_I + V_{II} + \varepsilon_n(\mathbf{R})) \phi_n(\mathbf{R}) = E \phi_n(\mathbf{R})$$

"adiabatic"

0. Structure of matter

Adiabatic approximation (Born-Oppenheimer approximation)

$$(T_e + V_{ee} + V_{eI}) \psi_n(\mathbf{r}, \mathbf{R}) = \varepsilon_n(\mathbf{R}) \psi_n(\mathbf{r}, \mathbf{R}) \quad ; \quad \tilde{H}_e = T_e + V_{ee} + V_{eI}$$



electronic band structure

$$(T_I + V_{II} + \varepsilon_n(\mathbf{R})) \phi_n(\mathbf{R}) = E \phi_n(\mathbf{R})$$



nucleic (phonon) state

0. Structure of matter

Classical nucleic motion

Ehrenfest theorem :

$$i\hbar \frac{\partial}{\partial t} \langle \mathbf{R} \rangle = \langle [H, \mathbf{R}] \rangle = i\hbar \frac{\langle \mathbf{P} \rangle}{M}, \quad i\hbar \frac{\partial}{\partial t} \langle \mathbf{P} \rangle = \langle [H, \mathbf{P}] \rangle = -i\hbar \langle \nabla \varepsilon_n(\mathbf{R}) \rangle - i\hbar \nabla V_{II}$$

$$M \frac{\partial^2}{\partial t^2} \langle \mathbf{R} \rangle = -\langle \nabla \varepsilon_n(\mathbf{R}) \rangle - \nabla V_{II}$$

$$\langle \mathbf{R} \rangle \rightarrow \mathbf{R}(t), \quad \underbrace{\langle \nabla \varepsilon_n(\mathbf{R}) \rangle}_{\text{Hellmann-Feynman theorem}} \rightarrow \nabla \varepsilon_n(\mathbf{R})$$

• Hellmann-Feynman theorem

$$\frac{\partial}{\partial \mathbf{R}_I} \varepsilon_n(\mathbf{R}) = \left\langle \psi_n \left| \frac{\partial}{\partial \mathbf{R}_I} \tilde{H}_e(\mathbf{R}) \right| \psi_n \right\rangle$$

$$M \frac{\partial^2}{\partial t^2} \mathbf{R}_I(t) = -\nabla_I \varepsilon_n(\mathbf{R}) - \nabla_I V_{II}$$

0. Structure of matter
1. Many-body theory of electron system
2. Density functional theory
3. Exchange and correlation in DFT
4. Practical solving of electronic problem

1. Many-body theory of electron system

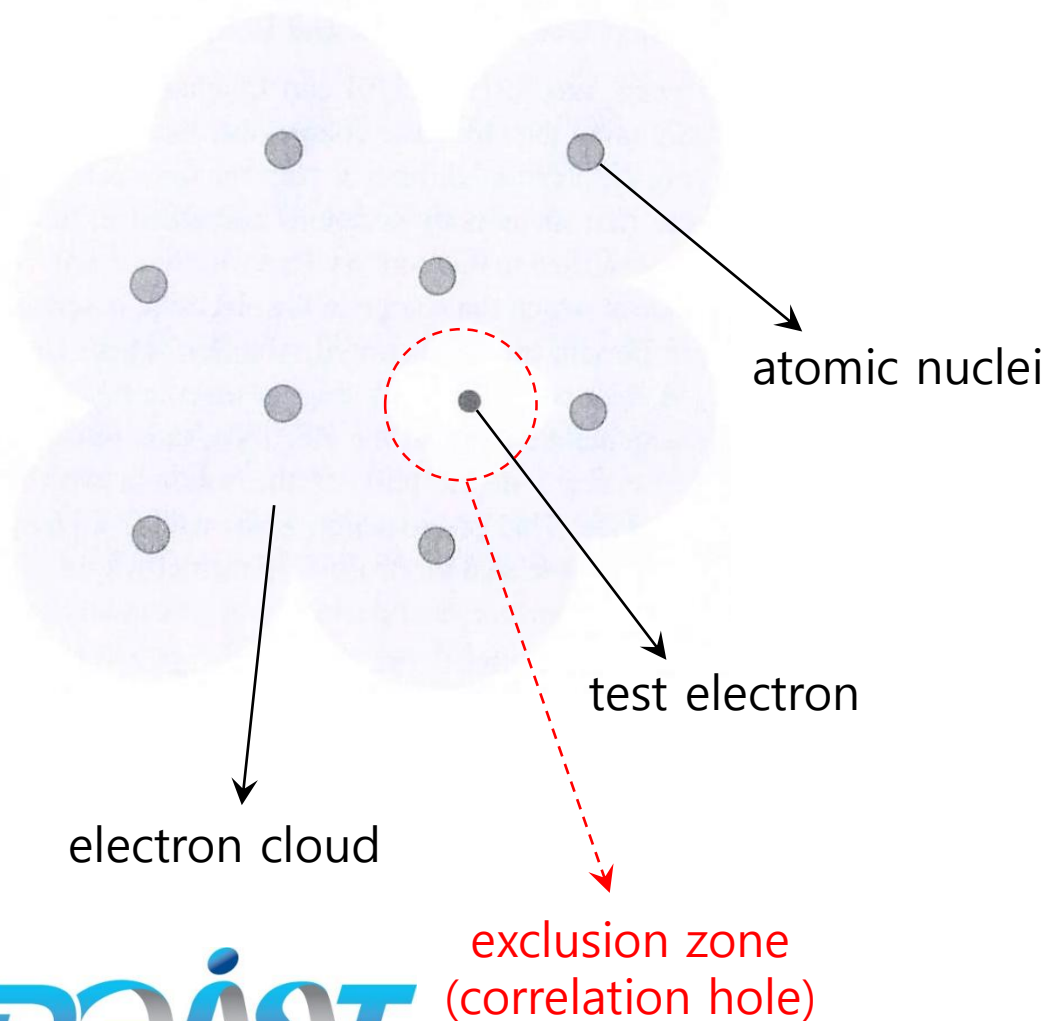
Fundamental Hamiltonian

$$H\psi(\mathbf{r}, \mathbf{R}) = \varepsilon(\mathbf{R})\psi(\mathbf{r}, \mathbf{R})$$

$$\begin{aligned} H &= T_e + V_{ee} + V_{eI} \\ &= -\frac{\hbar^2}{2m} \sum_i \nabla_i^2 + \frac{e^2}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} - e^2 \sum_{i,I} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} \end{aligned}$$

1. Many-body theory of electron system

Pair correlation function



$$\nabla^2 v(\mathbf{r}) = 4\pi[-e\delta(\mathbf{r} - \mathbf{r}_0) - en(\mathbf{r}, \mathbf{r}_0) + en]$$

; $\mathbf{r}_0$: location of test electron

- $n(\mathbf{r}, \mathbf{r}_0)$ depends on the location of test electron
: *test electron discourages other electrons from approaching it due to the Coulomb repulsion*
"correlation"
- pair correlation function is defined as $g(\mathbf{r}, \mathbf{r}_0) = n(\mathbf{r}, \mathbf{r}_0)/n$
: *pair correlation function represents the probability of finding an electron at $\mathbf{r}$ given that there is another electron at $\mathbf{r}_0$*

1. Many-body theory of electron system

Pair correlation function

$$\nabla^2 v(\mathbf{r}) = -4\pi e [\delta(\mathbf{r} - \mathbf{r}_0) + \underbrace{n\{g(\mathbf{r}, \mathbf{r}_0) - 1\}}]$$

essence of many-body problem

→ finding appropriate solution for
the pair correlation function

1. Many-body theory of electron system

Pair correlation function

$$\begin{aligned} E &= \langle \Psi | (T + V_{ext} + V_{ee}) | \Psi \rangle \\ &= \langle \Psi | T | \Psi \rangle + \langle \Psi | V_{ext} | \Psi \rangle + \langle \Psi | V_{ee} | \Psi \rangle \end{aligned}$$

; E : ground state energy
; $|\Psi\rangle$: N -electron ground state
; V_{ext} : external potential, i.e., V_{el}

one-body density matrix

$$\rho_1(\mathbf{r}, \mathbf{r}') = \sum_{\sigma} \langle \Psi | \varphi_{\sigma}^{\dagger}(\mathbf{r}) \varphi_{\sigma}(\mathbf{r}') | \Psi \rangle$$

$$; \{ \varphi_{\sigma}^{\dagger}(\mathbf{r}), \varphi_{\sigma'}(\mathbf{r}') \} = \delta_{\sigma\sigma'} \delta(\mathbf{r} - \mathbf{r}')$$

two-body density matrix

: fermionic field operator

$$\rho_2(\mathbf{r}, \mathbf{r}') = \sum_{\sigma\sigma'} \langle \Psi | \varphi_{\sigma}^{\dagger}(\mathbf{r}) \varphi_{\sigma'}^{\dagger}(\mathbf{r}') \varphi_{\sigma'}(\mathbf{r}') \varphi_{\sigma}(\mathbf{r}) | \Psi \rangle$$

$$\rho_2(\mathbf{r}, \mathbf{r}') = \rho_1(\mathbf{r}, \mathbf{r}) \rho_1(\mathbf{r}', \mathbf{r}') g(\mathbf{r}, \mathbf{r}')$$

$$; \rho_1(\mathbf{r}, \mathbf{r}) = \rho(\mathbf{r})$$

1. Many-body theory of electron system

Pair correlation function

$$\langle T \rangle = \langle \Psi | T | \Psi \rangle = -\frac{\hbar^2}{2m} \int \nabla^2 \rho(\mathbf{r}) d\mathbf{r} \quad ; \quad \rho(\mathbf{r}) = \rho_1(\mathbf{r}, \mathbf{r})$$

$$\langle V_{ext} \rangle = \langle \Psi | V_{ext} | \Psi \rangle = \int \rho(\mathbf{r}) V_{ext}(\mathbf{r}) d\mathbf{r}$$

$$; \quad V_{ext} = V_{eI} = \sum_i \sum_I v_{eI}(\mathbf{r}_i - \mathbf{R}_I) \rightarrow \langle V_{ext} \rangle = \sum_I \int \rho(\mathbf{r}) v_{eI}(\mathbf{r} - \mathbf{R}_I) d\mathbf{r}$$

$$\langle V_{ee} \rangle = \langle \Psi | V_{ee} | \Psi \rangle = \frac{1}{2} \int \int \frac{\rho_2(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$

$$T = -\frac{\hbar^2}{2m} \sum_{\sigma} \int dr \, \varphi_{\sigma}^{\dagger}(\mathbf{r}) \nabla^2 \varphi_{\sigma}(\mathbf{r}) = -\frac{\hbar^2}{2m} \sum_{\sigma} \int dr \, \nabla^2 \varphi_{\sigma}^{\dagger}(\mathbf{r}) \varphi_{\sigma}(\mathbf{r})$$

$$\langle \Psi | T | \Psi \rangle = -\frac{\hbar^2}{2m} \int \nabla^2 \rho(\mathbf{r}) d\mathbf{r}$$

$$V_{ee} = \frac{1}{2} \sum_{\sigma\sigma'} \int dr \int dr' \, \varphi_{\sigma}^{\dagger}(\mathbf{r}) \varphi_{\sigma'}^{\dagger}(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \varphi_{\sigma'}(\mathbf{r}') \varphi_{\sigma}(\mathbf{r})$$

$$\langle \Psi | V_{ee} | \Psi \rangle = \frac{1}{2} \int \int \frac{\rho_2(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$

1. Many-body theory of electron system

Pair correlation function

$$\langle V_{ee} \rangle = \langle \Psi | V_{ee} | \Psi \rangle = \frac{1}{2} \int \int \frac{\rho_2(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad ; \quad \rho_2(\mathbf{r}, \mathbf{r}') = \rho_1(\mathbf{r}, \mathbf{r}) \rho_1(\mathbf{r}', \mathbf{r}') g(\mathbf{r}, \mathbf{r}') = \rho(\mathbf{r}) \rho(\mathbf{r}') g(\mathbf{r}, \mathbf{r}')$$

$$= \underbrace{\frac{1}{2} \int \int \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'}_{\text{Hartree term}} + \underbrace{\frac{1}{2} \int \int \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} [g(\mathbf{r}, \mathbf{r}') - 1] d\mathbf{r} d\mathbf{r}'}_{\text{exchange and correlation effect}}$$

- classical electrostatic interaction for charge density $\rho(\mathbf{r})$
- called the *Hartree term*

- $g(\mathbf{r}, \mathbf{r}')$ implies that an electron at $\mathbf{r}$ discourages other electrons at $\mathbf{r}'$ from being close to $\mathbf{r}$ due to the Coulomb repulsion
- *exchange and correlation effect*
- classically, $g(\mathbf{r}, \mathbf{r}') \sim e^{-v(\mathbf{r}-\mathbf{r}')/k_B T}$

1. Many-body theory of electron system

Exchange and correlation

Exchange (Hartree-Fock theory) :

"correlation neglected, only the Pauli exclusion principle considered"

$$\langle V_{ee}^{\text{HF}} \rangle = \frac{1}{2} \int \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}d\mathbf{r}' + \frac{1}{2} \int \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} [g_X(\mathbf{r}, \mathbf{r}') - 1] d\mathbf{r}d\mathbf{r}'$$

$$g_X(\mathbf{r}, \mathbf{r}') \rightarrow \frac{1}{2}, \quad \text{for } \mathbf{r} \rightarrow \mathbf{r}'$$

$$\begin{aligned} g_{\uparrow\uparrow}(\mathbf{r}, \mathbf{r}') &\rightarrow 0 \\ g_{\uparrow\downarrow}(\mathbf{r}, \mathbf{r}') &\rightarrow 1 \end{aligned} \quad \text{for } \mathbf{r} \rightarrow \mathbf{r}'$$
$$g_X(\mathbf{r}, \mathbf{r}') = \frac{1}{2} [g_{\uparrow\uparrow}(\mathbf{r}, \mathbf{r}') + g_{\uparrow\downarrow}(\mathbf{r}, \mathbf{r}')]$$

(unpolarized case)

$$g_X(\mathbf{r}, \mathbf{r}') = 1 - \frac{\sum_{\sigma} |\rho_{\sigma}(\mathbf{r}, \mathbf{r}')|^2}{\rho(\mathbf{r})\rho(\mathbf{r}')} \quad \text{"exchange hole"}$$

$$\rho_2(\mathbf{r}, \mathbf{r}') = \rho(\mathbf{r})\rho(\mathbf{r}')g(\mathbf{r}, \mathbf{r}')$$

$$= \rho(\mathbf{r})\rho(\mathbf{r}') - |\rho_1(\mathbf{r}, \mathbf{r}')|^2 \quad ; \text{ from Slater determinant}$$

$$g(\mathbf{r}, \mathbf{r}') = 1 - \frac{|\rho_1(\mathbf{r}, \mathbf{r}')|^2}{\rho(\mathbf{r})\rho(\mathbf{r}')}$$

$$; \quad |\rho_1(\mathbf{r}, \mathbf{r}')|^2 = \sum_{\sigma} |\rho_{\sigma}(\mathbf{r}, \mathbf{r}')|^2$$

1. Many-body theory of electron system

Exchange and correlation

N -particle wave function, i.e., Slater determinant

$$\Psi_{\lambda_1 \lambda_2 \dots \lambda_N}(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_{\lambda_1}(\mathbf{r}_1) & \dots & \phi_{\lambda_1}(\mathbf{r}_N) \\ \vdots & \ddots & \vdots \\ \phi_{\lambda_N}(\mathbf{r}_1) & \dots & \phi_{\lambda_N}(\mathbf{r}_N) \end{vmatrix}$$

$$g_{\lambda_1 \lambda_2}(\mathbf{r}_1, \mathbf{r}_2) = V^2 \int d\mathbf{r}_3 d\mathbf{r}_4 \dots d\mathbf{r}_N \sum_{\lambda_3 \lambda_4 \dots \lambda_N} |\Psi_{\lambda_1 \lambda_2 \dots \lambda_N}(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)|^2 = \frac{V^2}{N(N-1)} \begin{vmatrix} \phi_{\lambda_1}(\mathbf{r}_1) & \phi_{\lambda_1}(\mathbf{r}_2) \\ \phi_{\lambda_2}(\mathbf{r}_1) & \phi_{\lambda_2}(\mathbf{r}_2) \end{vmatrix}^2$$

$$\phi_{\lambda}(\mathbf{r}) = \frac{\chi_{\sigma}}{\sqrt{V}} e^{i\mathbf{k} \cdot \mathbf{r}}; \quad g_{\uparrow\downarrow}(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{N(N-1)} \sum_{\mathbf{k}_1 \mathbf{k}_2} \left(|e^{i\mathbf{k}_1 \cdot \mathbf{r}_1 + i\mathbf{k}_2 \cdot \mathbf{r}_2}|^2 + |e^{i\mathbf{k}_1 \cdot \mathbf{r}_2 + i\mathbf{k}_2 \cdot \mathbf{r}_1}|^2 \right) = \frac{1}{2}$$

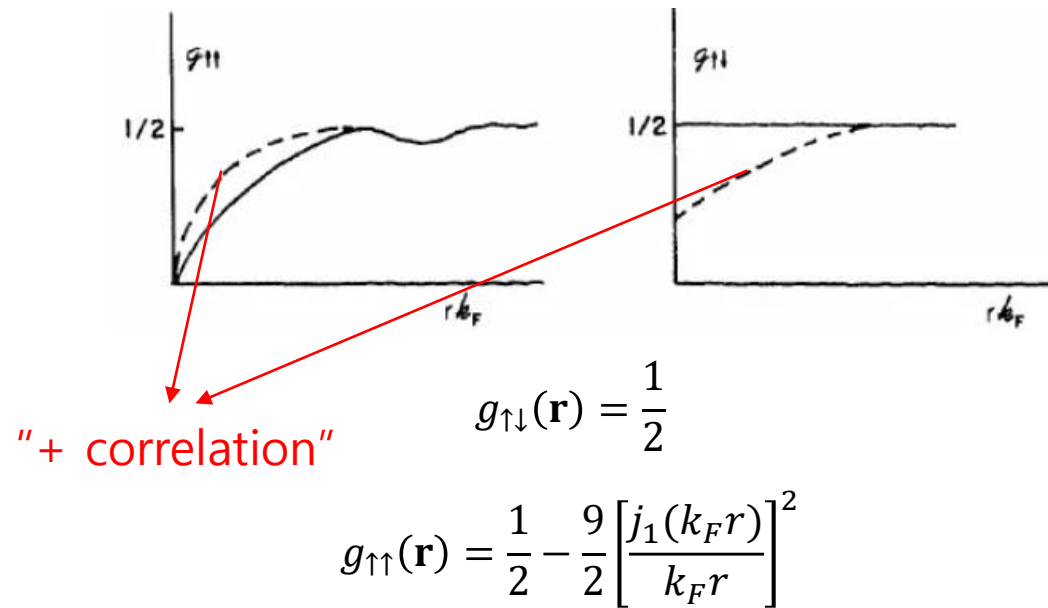
(homogeneous
electron gas)

$$\begin{aligned} g_{\uparrow\uparrow}(\mathbf{r}_1, \mathbf{r}_2) &= \frac{1}{N(N-1)} \sum_{\mathbf{k}_1 \mathbf{k}_2} |e^{i\mathbf{k}_1 \cdot \mathbf{r}_1 + i\mathbf{k}_2 \cdot \mathbf{r}_2} - e^{i\mathbf{k}_1 \cdot \mathbf{r}_2 + i\mathbf{k}_2 \cdot \mathbf{r}_1}|^2 \\ &= \frac{2}{N(N-1)} \sum_{\mathbf{k}_1 \mathbf{k}_2} [1 - e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot (\mathbf{r}_1 - \mathbf{r}_2)}] = \frac{1}{2} [1 - \Lambda(\mathbf{r}_1 - \mathbf{r}_2)^2] \end{aligned}$$

$$; \Lambda(\mathbf{r}) = \frac{2}{N} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{2}{n_0} \int \frac{d\mathbf{k}}{(2\pi)^3} n_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{3}{rk_F^3} \int_0^{k_F} k dk \sin kr = \frac{3}{rk_F} j_1(k_F r)$$

1. Many-body theory of electron system

Exchange and correlation



$$g_X(\mathbf{r}) = g_{\uparrow\uparrow}(\mathbf{r}) + g_{\uparrow\downarrow}(\mathbf{r})$$
$$= 1 - \frac{9}{2} \left[\frac{j_1(k_F r)}{k_F r} \right]^2$$

1. Many-body theory of electron system

Exchange and correlation

- $$\int d\mathbf{r} \rho(\mathbf{r}) [g_X(\mathbf{r}) - 1] = n_0 \int d\mathbf{r} [g_X(\mathbf{r}) - 1] = -1$$
$$\rightarrow \frac{9n_0}{2} \int d\mathbf{r} \left[\frac{j_1(k_F r)}{k_F r} \right]^2 = -\frac{6}{\pi} \int_0^\infty dx [j_1(x)]^2 = -1$$

normalization sum rule

- *exchange hole (or exchange-correlation hole) contains exactly one displaced electron*

- $$E_X = \frac{e^2 n_0}{2} \int d\mathbf{r} \frac{g_X(r) - 1}{r}$$
$$\rightarrow -\frac{9e^2 n_0}{4} \int d\mathbf{r} \frac{1}{r} \left[\frac{j_1(k_F r)}{k_F r} \right]^2 = -\frac{3}{\pi} e^2 n_0 \int_0^\infty dx \frac{[j_1(x)]^2}{x}$$
$$\rightarrow E_X = -\frac{3e^2 k_F}{4\pi}$$

exchange energy given in density

1. Many-body theory of electron system

Exchange and correlation

exchange energy - homogeneous electron gas

$$\begin{aligned}\langle V_{ee} \rangle &= \frac{1}{2V} \sum_{\mathbf{q}} \sum_{\mathbf{k}\mathbf{k}'} \sum_{\sigma\sigma'} v_{\mathbf{q}} \left\langle c_{\mathbf{k}+\mathbf{q}\sigma}^\dagger c_{\mathbf{k}'-\mathbf{q}\sigma'}^\dagger c_{\mathbf{k}'\sigma'} c_{\mathbf{k}\sigma} \right\rangle \quad ; v_{\mathbf{q}} = \frac{4\pi e^2}{\mathbf{q}^2} \\ &\approx \frac{1}{2V} \sum_{\mathbf{q}} \sum_{\mathbf{k}\mathbf{k}'} \sum_{\sigma\sigma'} v_{\mathbf{q}} \left[\underbrace{\langle c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} \rangle \langle c_{\mathbf{k}'\sigma'}^\dagger c_{\mathbf{k}'\sigma'} \rangle \delta_{\mathbf{q}=0}}_{\text{Hartree energy}} - \underbrace{\langle c_{\mathbf{k}+\mathbf{q}\sigma}^\dagger c_{\mathbf{k}'\sigma'} \rangle \langle c_{\mathbf{k}'-\mathbf{q}\sigma'}^\dagger c_{\mathbf{k}\sigma} \rangle}_{\text{Hartree-Fock exchange } E_X} \right]\end{aligned}$$

- cancelled by uniform positive background

$$E_X = \frac{1}{2N} \sum_{\mathbf{k}\sigma} n_{\mathbf{k}} \Sigma_X(\mathbf{k}), \quad \Sigma_X(\mathbf{k}) = -\frac{1}{V} \sum_{\mathbf{q}} v_{\mathbf{q}} n_{\mathbf{k}+\mathbf{q}}$$

$$\Sigma_X(\mathbf{k}) = \text{---}\overbrace{\hspace{1cm}}^{\hspace{1cm}}\text{---}$$

1. Many-body theory of electron system

Exchange and correlation

$$\begin{aligned}\Sigma_X(\mathbf{k}) &= -\frac{1}{(2\pi)^3} \int d\mathbf{p} \frac{4\pi e^2}{|\mathbf{p} - \mathbf{k}|^2} n_{\mathbf{p}} = -\frac{e^2}{\pi} \int_0^{k_F} dp p^2 \int_{-1}^1 \frac{dv}{k^2 + p^2 - 2pkv} \\ &= -\frac{e^2 k_F}{\pi} \left(1 + \frac{1-y^2}{2y} \ln \left| \frac{1+y}{1-y} \right| \right) \quad ; \quad y = k/k_F\end{aligned}$$

$$\begin{aligned}E_X &= \frac{1}{2N} \sum_{\mathbf{k}\sigma} n_{\mathbf{k}} \Sigma_X(\mathbf{k}) = \frac{1}{n_0} \frac{1}{(2\pi)^3} \int d\mathbf{k} n_{\mathbf{k}} \Sigma_X(\mathbf{k}) \\ &= -\frac{k_F^3}{2\pi^2 n_0} \frac{e^2 k_F}{\pi} \int_0^1 dy y^2 \left(1 + \frac{1-y^2}{2y} \ln \left| \frac{1+y}{1-y} \right| \right) \\ &= -\frac{3}{4} \frac{e^2 k_F}{\pi} = -\frac{0.458}{r_s} \quad \rightarrow \quad E_X[\rho] = -\frac{3}{4} \left(\frac{3}{\pi} \right)^{1/3} \int d\mathbf{r} \rho(\mathbf{r})^{4/3}\end{aligned}$$

$$; \quad r_s = \left(\frac{3}{4\pi n_0} \right)^{1/3}$$

in Thomas-Fermi-Dirac (TFD) theory

1. Many-body theory of electron system

Exchange and correlation

Digression – homogeneous electron gas

$$\frac{4\pi n_0}{3} r_s^3 = 1 \quad ; \quad r_s : \text{in unit of Bohr radius}$$

$$n_0 = 2 \sum_{\mathbf{k}} n_{\mathbf{k}} = \frac{1}{\pi^2} \int_0^{k_F} dk k^2 = \frac{k_F^3}{3\pi^2}, \quad k_F = (3\pi^2 n_0)^{1/3}$$

$$\text{Fermi energy : } E_F = \frac{\hbar^2 k_F^2}{2m}$$

$$\text{Kinetic energy per electron : } \langle T \rangle = \frac{2}{N} \sum_{\mathbf{k}} n_{\mathbf{k}} \varepsilon_{\mathbf{k}} = \frac{3}{5} \frac{\hbar^2 k_F^2}{2m} = \frac{3}{5} E_F = \frac{1.105}{r_s^2}$$

$$\rightarrow T[\rho] = \frac{3}{10} (3\pi^2)^{1/3} \int d\mathbf{r} \rho(\mathbf{r})^{5/3} \quad \text{in TFD theory}$$

1. Many-body theory of electron system

Exchange and correlation

Energy of many-body electron system :

$$E = \langle T \rangle + \langle V_{ext} \rangle + \frac{1}{2} \int \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}d\mathbf{r}' + E_{XC}$$

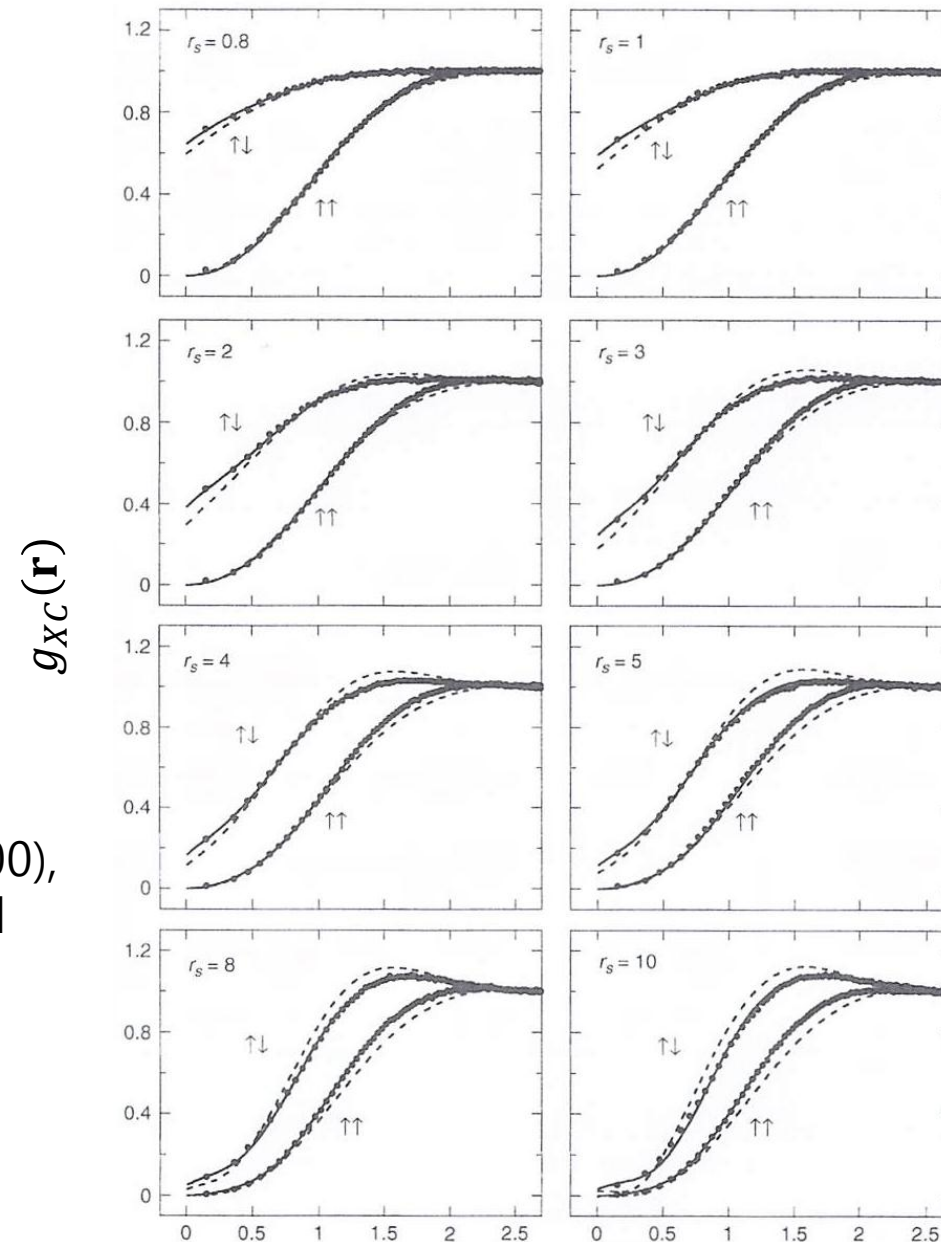
$$E_{XC} = \frac{1}{2} \int \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} [g_{XC}(\mathbf{r}, \mathbf{r}') - 1] d\mathbf{r}d\mathbf{r}'$$

$$g_{XC}(\mathbf{r}, \mathbf{r}') = g_X(\mathbf{r}, \mathbf{r}') + \underbrace{g_C(\mathbf{r}, \mathbf{r}')}_{\text{"correlation hole"}}$$

"correlation hole"

1. Many-body theory of electron system

Exchange and correlation



dashed line : Phys. Rev. B 61, 7353 (2000),
from Perdew-Wang model

dots : Phys. Rev. Lett. 82, 5317 (1999),
from QMC calculation

Perdew-Wang model

$$g_{PW} = g_X + g_C$$

$$g_X = 1 - \frac{9}{2} \left[\frac{j_1(k_F r)}{k_F r} \right]^2$$

$$g_C = g_C(r_s, \zeta, r) \quad ; \quad \zeta = \frac{n_\uparrow - n_\downarrow}{n}$$

$$= A(r_s)F(r) + B(r_s)G(r) + \dots$$

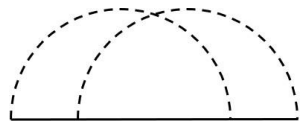
; interpolating various limits

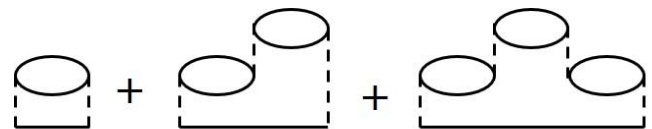
(normalization, long-range behavior, reproduction of QMC, ...)

1. Many-body theory of electron system

Exchange and correlation

$$E_C = A + B \ln r_s + \dots \quad ; \text{high density limit } (r_s < 1)$$


$$= \frac{1}{6} \ln 2 - \frac{3}{4\pi^2} \zeta(3) = 0.0218$$


$$+ \dots = \Sigma_{RPA} \approx -0.071 + 0.0311 \ln r_s + \dots$$

}

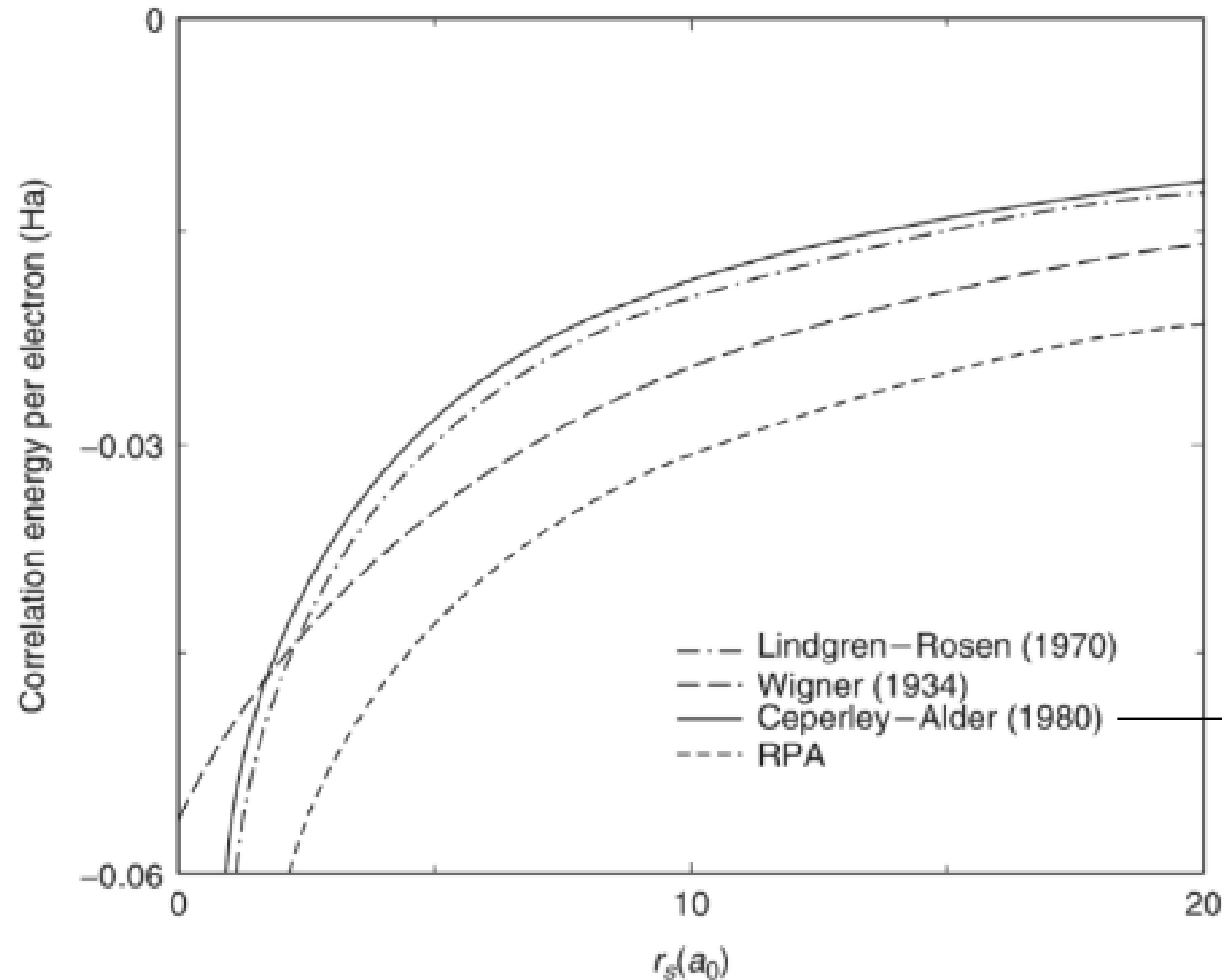
$\approx -0.049 + 0.0311 \ln r_s$
"Gell-Mann – Brueckner formula"

$$E_C = \frac{0.44}{r_s + 7.8} \quad ; \text{low density limit } (r_s > 1)$$

"Wigner formula"

1. Many-body theory of electron system

Exchange and correlation



QMC results

1. Many-body theory of electron system

Hartree and Hartree-Fock equation

Hartree equation :

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \phi_1(\mathbf{r}_1)\phi_2(\mathbf{r}_2) \cdots \phi_N(\mathbf{r}_N)$$

$$E = \langle \Psi | H | \Psi \rangle$$

$$= \langle \Psi | T | \Psi \rangle + \langle \Psi | V_{ext} | \Psi \rangle + \langle \Psi | V_{ee} | \Psi \rangle$$

$$= \sum_i \int d\mathbf{r} \phi_i^*(\mathbf{r}) \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) \right] \phi_i(\mathbf{r}) + \frac{1}{2} \sum_{i \neq j} \int \int d\mathbf{r} d\mathbf{r}' \phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_i(\mathbf{r}) \phi_j(\mathbf{r}')$$

$$\frac{\delta}{\delta \phi_i^*} \left(E[\phi_i] - \sum_i \epsilon_i (\langle \phi_i | \phi_i \rangle - 1) \right) = 0$$

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) + \sum_{j(\neq i)} \frac{e^2 |\phi_j(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} \right] \phi_i(\mathbf{r}) = \epsilon_i \phi_i(\mathbf{r})$$

1. Many-body theory of electron system

Hartree and Hartree-Fock equation

Hartree-Fock equation :

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(\mathbf{r}_1) & \cdots & \phi_1(\mathbf{r}_N) \\ \vdots & \ddots & \vdots \\ \phi_N(\mathbf{r}_1) & \cdots & \phi_N(\mathbf{r}_N) \end{vmatrix}$$

$$E = \langle \Psi | H | \Psi \rangle$$

$$= \langle \Psi | T | \Psi \rangle + \langle \Psi | V_{ext} | \Psi \rangle + \langle \Psi | V_{ee} | \Psi \rangle$$

$$= \sum_i \int d\mathbf{r} \phi_i^*(\mathbf{r}) \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) \right] \phi_i(\mathbf{r}) + \frac{1}{2} \sum_{i \neq j} \int \int d\mathbf{r} d\mathbf{r}' \phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_i(\mathbf{r}) \phi_j(\mathbf{r}') \\ - \frac{1}{2} \sum_{i \neq j} \int \int d\mathbf{r} d\mathbf{r}' \phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_i(\mathbf{r}') \phi_j(\mathbf{r}) \delta_{\sigma_i \sigma_j}$$

$$\frac{\delta}{\delta \phi_i^*} \left(E[\phi_i] - \sum_i \epsilon_i (\langle \phi_i | \phi_i \rangle - 1) \right) = 0$$

1. Many-body theory of electron system

Hartree and Hartree-Fock equation

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) + \sum_{j(\neq i)} \frac{e^2 |\phi_j(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} \right] \phi_i(\mathbf{r}) - \underbrace{\sum_j \int d\mathbf{r}' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_j^*(\mathbf{r}') \phi_i(\mathbf{r}') \phi_j(\mathbf{r}) \delta_{\sigma_i \sigma_j}}_{\text{exchange potential part} \rightarrow \text{nonlocal}} = \epsilon_i \phi_i(\mathbf{r})$$

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) + \sum_{j(\neq i)} \frac{e^2 |\phi_j(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} + V_{ex}(\mathbf{r}) \delta_{\sigma_i \sigma_j} \right] \phi_i(\mathbf{r}) = \epsilon_i \phi_i(\mathbf{r})$$

1. Many-body theory of electron system

Hartree and Hartree-Fock equation

$$V_{ex}(\mathbf{r}) = - \sum_j \int d\mathbf{r}' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \phi_j^*(\mathbf{r}') \phi_i(\mathbf{r}') \frac{\phi_j(\mathbf{r})}{\phi_i(\mathbf{r})} = - \int d\mathbf{r}' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \underbrace{\sum_j \frac{\phi_j^*(\mathbf{r}') \phi_i(\mathbf{r}') \phi_j(\mathbf{r}) \phi_i^*(\mathbf{r})}{\phi_i^*(\mathbf{r}) \phi_i(\mathbf{r})}}$$

→ like exchange hole

$$\sum_j \frac{\phi_j^*(\mathbf{r}') \phi_i(\mathbf{r}') \phi_j(\mathbf{r}) \phi_i^*(\mathbf{r})}{\phi_i^*(\mathbf{r}) \phi_i(\mathbf{r})} \approx \frac{\sum_{ij} \phi_j^*(\mathbf{r}') \phi_i(\mathbf{r}') \phi_j(\mathbf{r}) \phi_i^*(\mathbf{r})}{\sum_i \phi_i^*(\mathbf{r}) \phi_i(\mathbf{r})} \quad ; \quad \phi_i(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}_i \cdot \mathbf{r}}$$

$$\approx \frac{\sum_{ij} e^{-i\mathbf{k}_j \cdot \mathbf{r}'} e^{i\mathbf{k}_i \cdot \mathbf{r}'} e^{i\mathbf{k}_j \cdot \mathbf{r}} e^{-i\mathbf{k}_i \cdot \mathbf{r}}}{\sum_i e^{-i\mathbf{k}_i \cdot \mathbf{r}} e^{i\mathbf{k}_i \cdot \mathbf{r}}}$$

$$= \frac{\sum_{ij} e^{-i(\mathbf{k}_i - \mathbf{k}_j) \cdot (\mathbf{r} - \mathbf{r}')}}{\sum_i e^{-i\mathbf{k}_i \cdot \mathbf{r}} e^{i\mathbf{k}_i \cdot \mathbf{r}}} = -n_0 [g_X(\mathbf{r} - \mathbf{r}') - 1]$$

$$= \frac{9}{2} n_0 \left[\frac{j_1(k_F |\mathbf{r} - \mathbf{r}'|)}{k_F |\mathbf{r} - \mathbf{r}'|} \right]^2$$

1. Many-body theory of electron system

Hartree and Hartree-Fock equation

$$\begin{aligned} V_{ex}(\mathbf{r}) &= -n_0 \int d\mathbf{r}' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} [g_X(\mathbf{r} - \mathbf{r}') - 1] = -\frac{3e^2 k_F}{2\pi} = -\frac{3}{2} \left(\frac{3}{\pi} \right)^{1/3} n_0^{1/3} \\ &= -\frac{3}{2} \left(\frac{3}{\pi} \right)^{1/3} \rho(\mathbf{r})^{1/3} \end{aligned}$$

→ local exchange potential (Hartree-Fock-Slater model)

- X_α method

$$V_{ex}(\mathbf{r}) = -\frac{3}{2} \alpha \left(\frac{3}{\pi} \right)^{1/3} \rho(\mathbf{r})^{1/3}, \quad 0 < \alpha < 1$$

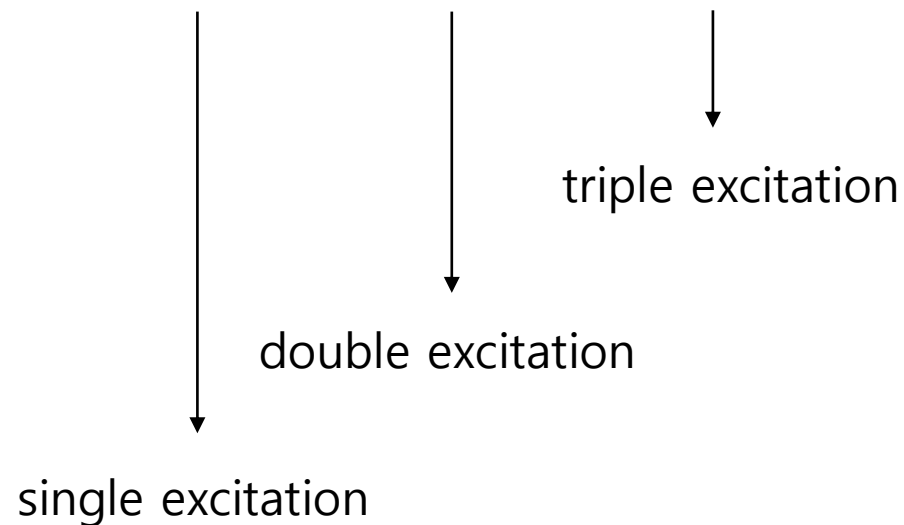
1. Many-body theory of electron system

Post Hartree-Fock method

$$E_C = E_{tot} - E_{HF}$$

Configuration interaction (CI) method :

$$\Psi_{CI} = C_0 \Psi_{HF} + \sum_{i_S} C_i^S \Psi_i^S + \sum_{i_D} C_i^D \Psi_i^D + \sum_{i_T} C_i^T \Psi_i^T + \dots$$



→ linear combination of Slater determinants

0. Structure of matter
1. Many-body theory of electron system
- 2. Density functional theory**
3. Exchange and correlation in DFT
4. Practical solving of electronic problem

2. Density functional theory

Thomas-Fermi theory

- describing energy of electron system in terms of the electronic density
- based on the homogeneous electron gas

$$E_\alpha[\rho] = \int d\mathbf{r} \rho(\mathbf{r}) \underbrace{\epsilon_\alpha[\rho]}_{\text{energy density of contribution } \alpha}$$

• kinetic energy : $\langle T \rangle = \frac{3}{5} \frac{\hbar^2 k_F^2}{2m} \quad \rightarrow \quad E_T[\rho] = \frac{3}{10} (3\pi^2)^{2/3} \int d\mathbf{r} \rho(\mathbf{r})^{5/3}$

$$= 2.871 \int d\mathbf{r} \rho(\mathbf{r})^{5/3}$$

• exchange energy : $E_X = -\frac{3}{4} \frac{e^2 k_F}{\pi} \quad \rightarrow \quad E_X[\rho] = \frac{3}{4} (3\pi^2)^{1/3} \int d\mathbf{r} \rho(\mathbf{r})^{4/3}$

$$= 0.739 \int d\mathbf{r} \rho(\mathbf{r})^{4/3}$$

(Thomas-Fermi-Dirac)

2. Density functional theory

Thomas-Fermi theory

- correlation :
$$E_C = -\frac{0.44}{r_s + 7.8} = -\frac{0.44n_0^{1/3}}{\left(\frac{3}{4\pi}\right)^{1/3} + 7.8n_0^{1/3}} \quad (\text{Wigner formula})$$

$$\rightarrow E_C[\rho] = -0.056 \int d\mathbf{r} \frac{\rho(\mathbf{r})^{4/3}}{0.079 + \rho(\mathbf{r})^{1/3}}$$

$$E_{TFD}[\rho] = 2.871 \int d\mathbf{r} \rho(\mathbf{r})^{5/3} + \int d\mathbf{r} \rho(\mathbf{r}) V_{ext}(\mathbf{r}) + \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \\ - 0.739 \int d\mathbf{r} \rho(\mathbf{r})^{4/3} - 0.056 \int d\mathbf{r} \frac{\rho(\mathbf{r})^{4/3}}{0.079 + \rho(\mathbf{r})^{1/3}}$$



$E_{TFD}[\rho]$ is a functional of the density

2. Density functional theory

Thomas-Fermi theory

Digression – functional derivatives

$$\int dx \frac{\delta F[y(x)]}{\delta y(x)} \phi(x) = \lim_{\epsilon \rightarrow 0} \frac{F[y(x) + \epsilon \phi(x)] - F[y(x)]}{\epsilon}$$

$$F[y(x)] = y(x) \rightarrow \frac{\delta}{\delta y(x)} \int dx F[y(x)] \phi(x) = \int dx \frac{\delta F[y(x)]}{\delta y(x)} \phi(x) = \phi(x) \rightarrow \frac{\delta y(x')}{\delta y(x)} = \delta(x' - x)$$

$$\frac{\delta}{\delta f(x)} \left(\int dx g(x) f(x) \right) = g(x), \quad \frac{\delta}{\delta f(x)} \left(\frac{1}{2} \int \int dx dx' g(x, x') f(x) f(x') \right) = \int dx' g(x, x') f(x')$$

$$\frac{\delta^2}{\delta f(x) \delta f(x')} \left(\frac{1}{2} \int \int dx dx' g(x, x') f(x) f(x') \right) = g(x, x')$$

2. Density functional theory

Thomas-Fermi theory

Thomas-Fermi-Dirac equation :

$$\frac{\delta}{\delta\rho(\mathbf{r})} \left(E_{TFD}[\rho] - \mu \left(\int d\mathbf{r} \rho(\mathbf{r}) - N \right) \right) = 0$$

$$\mu = 4.785\rho(\mathbf{r})^{2/3} + V_{ext}(\mathbf{r}) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} - 0.985\rho(\mathbf{r})^{1/3} + \frac{\delta E_c[\rho]}{\delta\rho(\mathbf{r})}$$

- thinking of the inhomogeneous system as locally homogeneous for the kinetic energy term
 - relatively reasonable for condensed phase, especially for simple metals
 - poor for atomic or molecular systems
 - better treatment of kinetic energy especially required

Thomas-Fermi theory : its implication

- electron density (distribution) could be obtained without solving Schrödinger equation
- external potential determines the electron density
 - similar to Hohenberg-Kohn theorem

2. Density functional theory

Kohn-Sham density functional theory

Hohenberg-Kohn theorem :

Theorem 1 (uniqueness)

- The external potential is univocally determined by the ground-state electronic density.*

proof : supposing the opposite to hold $\rightarrow$ two different external potential V_{ext} and V'_{ext}
for the same ground state electronic density ρ

$$E_0 = \langle \Psi | H | \Psi \rangle \quad \text{with} \quad H = T + V_{ext} + V_{ee} \quad , \quad E'_0 = \langle \Psi' | H' | \Psi' \rangle \quad \text{with} \quad H' = T + V'_{ext} + V_{ee}$$

$$E_0 < \langle \Psi' | H | \Psi' \rangle = \langle \Psi' | H' | \Psi' \rangle + \langle \Psi' | (H - H') | \Psi' \rangle = E'_0 + \int d\mathbf{r} \rho(\mathbf{r}) [V_{ext}(\mathbf{r}) - V'_{ext}(\mathbf{r})]$$

$$E'_0 < \langle \Psi | H' | \Psi \rangle = \langle \Psi | H | \Psi \rangle + \langle \Psi | (H' - H) | \Psi \rangle = E_0 - \int d\mathbf{r} \rho(\mathbf{r}) [V_{ext}(\mathbf{r}) - V'_{ext}(\mathbf{r})]$$

2. Density functional theory

Kohn-Sham density functional theory

Hohenberg-Kohn theorem :

Theorem 2

- The ground state energy is the minimum of the total energy at the correct ground state density ρ that satisfies $\int d\mathbf{r}\rho(\mathbf{r}) = N$.*

proof :

$$\begin{aligned}\langle \Psi[\tilde{\rho}] | H | \Psi[\tilde{\rho}] \rangle &= F[\tilde{\rho}] + \int d\mathbf{r} \tilde{\rho}(\mathbf{r}) V_{ext}(\mathbf{r}) \\ &= E[\tilde{\rho}] \geq E[\rho] = \langle \Psi[\rho] | H | \Psi[\rho] \rangle = F[\rho] + \int d\mathbf{r} \rho(\mathbf{r}) V_{ext}(\mathbf{r}) = E_0\end{aligned}$$

$$; F[\rho] = \langle \Psi[\rho] | (T + V_{ee}) | \Psi[\rho] \rangle$$

“universal functional”

$$\delta \left[E[\rho] - \mu \left(\int d\mathbf{r} \rho(\mathbf{r}) - N \right) \right] = 0$$

2. Density functional theory

Kohn-Sham density functional theory

Hohenberg-Kohn theorem and its true implication :

- to formulate the density functional theory as an *exact theory of many-body systems*
- information amount

$$\rho(\mathbf{r}) \rightarrow V_{ext} \rightarrow H \rightarrow \Psi \rightarrow \text{all observables}$$

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \longleftrightarrow \rho(\mathbf{r})$$

equivalent amount of information

2. Density functional theory

Kohn-Sham density functional theory

Levy constrained search :

$$F[\rho] = \min_{\Psi \rightarrow \rho} \langle \Psi[\rho] | (T + V_{ee}) | \Psi[\rho] \rangle$$

- out of Ψ to produce a certain density $\rho(\mathbf{r})$, we choose to minimize $\langle \Psi[\rho] | (T + V_{ee}) | \Psi[\rho] \rangle$
"necessarily exist"

$$\langle \Psi[\rho] | H | \Psi[\rho] \rangle = E[\rho] = F[\rho] + \int d\mathbf{r} \rho(\mathbf{r}) V_{ext}(\mathbf{r})$$

- for all possible density $\rho(\mathbf{r})$, we minimize $E[\rho]$
- a correct wave function of the density is guaranteed
- in a frame of Hohenberg-Kohn theorem, $\rho(\mathbf{r}) \rightarrow V_{ext}(r) \rightarrow \Psi$



2. Density functional theory

Kohn-Sham density functional theory

Kohn-Sham equation :

The equivalent noninteracting system, i.e., a system of noninteracting electrons whose ground state density coincides with that of the interacting system, does exist.

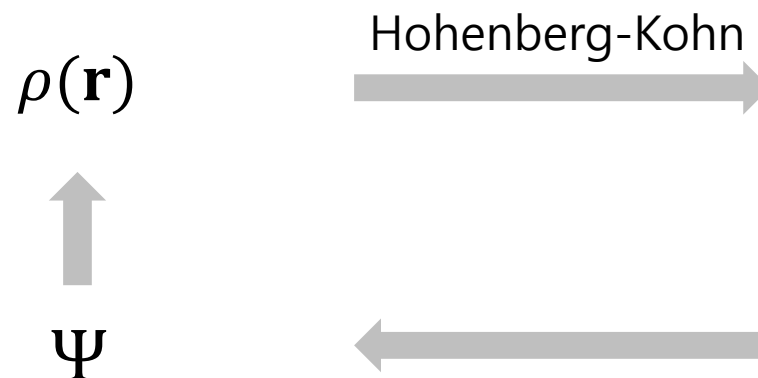
→ the noninteracting effective system of density $\rho(\mathbf{r})$



Walter Kohn



Lu Jeu Sham



$V_{eff}(\mathbf{r})$



$\phi_i(\mathbf{r})$

"orbital-based"

2. Density functional theory

Kohn-Sham density functional theory

$$H_{KS} = -\frac{\hbar^2}{2m}\nabla^2 + \underbrace{V_{eff}(\mathbf{r})}$$

- local effective (noninteracting) potential searching out the density of the interacting system

$$H_{KS}\phi_i(\mathbf{r}) = \varepsilon_i\phi_i(\mathbf{r})$$

- single-particle Kohn-Sham orbital
- an obvious meaning in constructing the density and kinetic energy
- physically meaningful, meaningless, or in between

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(\mathbf{r}_1) & \dots & \phi_1(\mathbf{r}_N) \\ \vdots & \ddots & \vdots \\ \phi_N(\mathbf{r}_1) & \dots & \phi_N(\mathbf{r}_N) \end{vmatrix}$$

$$\rho(\mathbf{r}) = \sum_i |\phi_i(\mathbf{r})|^2, \quad T_{eff}[\rho] = -\frac{\hbar^2}{2m} \sum_i \langle \phi_i | \nabla^2 | \phi_i \rangle$$

2. Density functional theory

Kohn-Sham density functional theory

Kohn and Sham's idea on kinetic energy correction (by introducing single-particle orbitals)

- for noninteracting electrons producing the same electronic density of the interacting system

$$T_{eff}[\rho] = -\frac{\hbar^2}{2m} \sum_i \langle \phi_i | \nabla^2 | \phi_i \rangle$$

- $T_{eff}[\rho]$ is NOT exact kinetic energy of the interacting system
- kinetic energy correction should be included in the correlation energy term

$$E_{xc}[\rho] + \text{kinetic energy correction} \rightarrow \tilde{E}_{xc}[\rho]$$

2. Density functional theory

Kohn-Sham density functional theory

$$F[\rho] = T_{eff}[\rho] + \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \tilde{E}_{xc}[\rho]$$

$$E_{KS}[\rho] = F[\rho] + \int d\mathbf{r} \rho(\mathbf{r}) V_{ext}(\mathbf{r}) = T_{eff}[\rho] + \int d\mathbf{r} \rho(\mathbf{r}) V_{ext}(\mathbf{r}) + \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \tilde{E}_{xc}[\rho]$$

- we should find $V_{eff}(\mathbf{r})$ that ensures the density of noninteracting Kohn-Sham system to be same as the true density of the interacting system
- minimizing the Kohn-Sham functional with respect to the density, under the constraint that the density integrates to N particles

2. Density functional theory

Kohn-Sham density functional theory

$$\frac{\delta}{\delta \rho(\mathbf{r})} \left[E_{KS}[\rho] - \mu \left(\int d\mathbf{r} \rho(\mathbf{r}) - N \right) \right] = 0$$

$$\underbrace{\frac{\delta T_{eff}[\rho]}{\delta \rho(\mathbf{r})} + V_{ext}(\mathbf{r}) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \frac{\delta \tilde{E}_{XC}[\rho]}{\delta \rho(\mathbf{r})}}_{= V_{eff}(\mathbf{r})} = \mu$$

- chemical potential of the noninteracting system, which should coincide with that of the interacting system

$$V_{eff}(\mathbf{r}) = V_{ext}(\mathbf{r}) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + V_{XC}([\rho]; \mathbf{r}) \quad ; \quad V_{XC}([\rho]; \mathbf{r}) = \frac{\delta \tilde{E}_{XC}[\rho]}{\delta \rho(\mathbf{r})}$$

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{eff}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r})$$

$$E_{V_{eff}}[\rho] = T_{eff}[\rho] + \int d\mathbf{r} \rho(\mathbf{r}) V_{eff}(\mathbf{r})$$

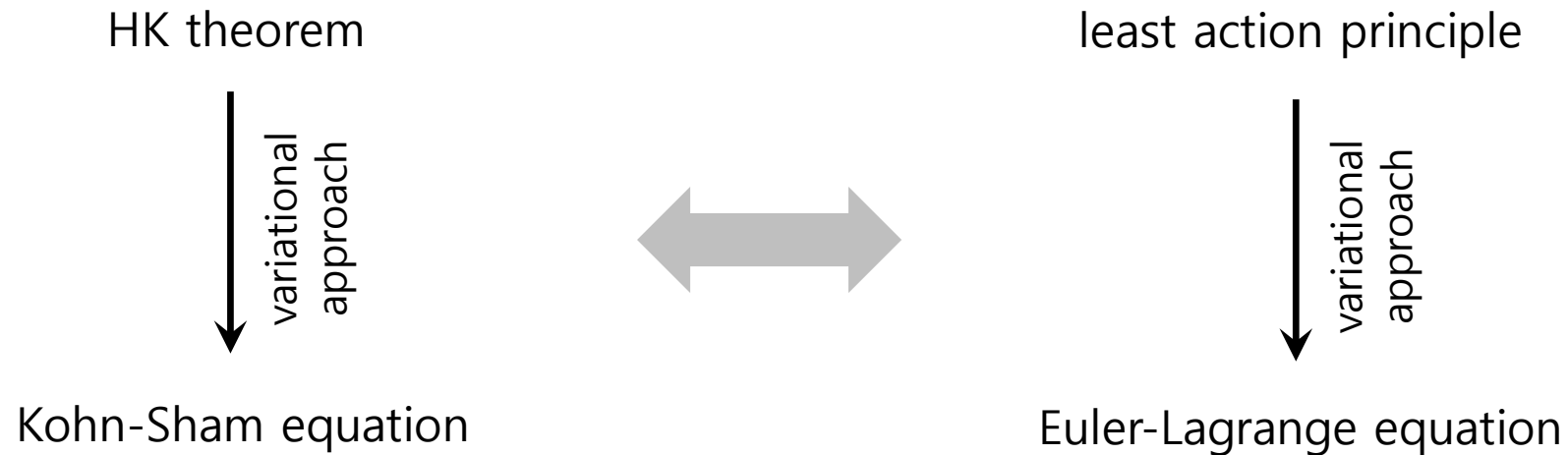
$$\frac{\delta}{\delta \rho(\mathbf{r})} \left[E_{KS}[\rho] - \mu \left(\int d\mathbf{r} \rho(\mathbf{r}) - N \right) \right] = 0 \quad \longleftrightarrow \quad \frac{\delta}{\delta \rho(\mathbf{r})} \left[E_{V_{eff}}[\rho] - \mu_{eff} \left(\int d\mathbf{r} \rho(\mathbf{r}) - N \right) \right] = 0$$

- $V_{eff}(\mathbf{r})$ ensures the density of noninteracting Kohn-Sham system to be same as the true density of the interacting system
- more exactly, $V_{eff}(\mathbf{r})$ is defined such that the density of noninteracting Kohn-Sham system should be same as the true density of the interacting system

2. Density functional theory

Kohn-Sham density functional theory

Hohenberg-Kohn theorem and Kohn-Sham equation :



2. Density functional theory

Kohn-Sham density functional theory

Spin polarized Kohn-Sham equation :

$$\rho(\mathbf{r}) = \rho_{\uparrow}(\mathbf{r}) + \rho_{\downarrow}(\mathbf{r}), \quad \zeta(\mathbf{r}) = \rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r}) \quad ; \quad \rho_{\sigma}(\mathbf{r}) = \sum_i |\phi_{i\sigma}(\mathbf{r})|^2$$

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{eff}^{\sigma}(\mathbf{r}) \right] \phi_{i\sigma}(\mathbf{r}) = \varepsilon_{i\sigma} \phi_{i\sigma}(\mathbf{r})$$

$$V_{eff}^{\sigma}(\mathbf{r}) = V_{ext}(\mathbf{r}) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + V_{XC}^{\sigma}([\rho_{\uparrow}, \rho_{\downarrow}]; \mathbf{r}) \quad ; \quad V_{XC}^{\sigma}([\rho_{\uparrow}, \rho_{\downarrow}]; \mathbf{r}) = \frac{\delta \tilde{E}_{XC}[\rho_{\uparrow}, \rho_{\downarrow}]}{\delta \rho_{\sigma}(\mathbf{r})}$$

$$E_{KS}[\rho_{\uparrow}, \rho_{\downarrow}] = T_{eff}[\rho_{\uparrow}, \rho_{\downarrow}] + \int d\mathbf{r} \rho(\mathbf{r}) V_{ext}(\mathbf{r}) + \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \tilde{E}_{XC}[\rho_{\uparrow}, \rho_{\downarrow}]$$

$$T_{eff}[\rho_{\uparrow}, \rho_{\downarrow}] = -\frac{\hbar^2}{2m} \sum_{i\sigma} \langle \phi_{i\sigma} | \nabla^2 | \phi_{i\sigma} \rangle$$

2. Density functional theory

Kohn-Sham density functional theory

Total energy calculation (double counting subtracted) :

① spin-unpolarized case

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{eff}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad V_{eff}(\mathbf{r}) = V_{ext}(\mathbf{r}) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + V_{XC}([\rho]; \mathbf{r})$$

$$E_{tot} = 2 \sum_i \varepsilon_i - \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \tilde{E}_{XC}[\rho] - \int d\mathbf{r} \rho(\mathbf{r}) V_{XC}([\rho]; \mathbf{r})$$

② spin-polarized case

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{eff}^\sigma(\mathbf{r}) \right] \phi_{i\sigma}(\mathbf{r}) = \varepsilon_{i\sigma} \phi_{i\sigma}(\mathbf{r}), \quad V_{eff}^\sigma(\mathbf{r}) = V_{ext}(\mathbf{r}) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + V_{XC}^\sigma([\rho_\uparrow, \rho_\downarrow]; \mathbf{r})$$

$$E_{tot} = \sum_{i\sigma} \varepsilon_{i\sigma} - \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \tilde{E}_{XC}[\rho_\uparrow, \rho_\downarrow] - \int d\mathbf{r} \rho(\mathbf{r}) V_{XC}^\sigma([\rho_\uparrow, \rho_\downarrow]; \mathbf{r})$$

2. Density functional theory

Kohn-Sham density functional theory

Kinetic energy correction (kinetic correlation) revisited :

$$T_{eff}[\rho] = \langle \Psi | T | \Psi \rangle \quad \leftarrow |\Psi\rangle : \text{Slater determinant of Kohn-Sham orbitals} \\ \text{(ground state of kinetic energy operator } T \text{)}$$

$$T[\rho] = \langle \Psi_0 | T | \Psi_0 \rangle \quad \leftarrow |\Psi_0\rangle : \text{true many-body ground state} \\ \text{(not a ground state of kinetic energy operator } T \text{)}$$

$$T_{eff}[\rho] \leq T[\rho]$$

$$\tilde{E}_{XC}[\rho] = E_{XC}[\rho] + \underbrace{T[\rho] - T_{eff}[\rho]}_{\geq 0}$$

$$\begin{array}{ccccc} H_0 & \rightarrow & H_0 + \lambda V_{ee} & \rightarrow & H_0 + V_{ee} \\ \lambda = 0 & \xrightarrow{\text{adiabatic connection}} & & & \lambda = 1 \end{array}$$

$$\tilde{g}(\mathbf{r}, \mathbf{r}') = \int_0^1 d\lambda g_\lambda(\mathbf{r}, \mathbf{r}')$$

※ Hellmann-Feynman theorem

$$\langle V_{ee} \rangle = \langle \Psi | V_{ee} | \Psi \rangle = \frac{1}{2} \int \int \frac{\rho_2(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' = \frac{1}{2} \int \int \frac{\rho(\mathbf{r}) \rho(\mathbf{r}') g(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$

$$\frac{\partial}{\partial \lambda} \langle \tilde{V}_{ee} \rangle = \langle V_{ee} \rangle_\lambda = \frac{1}{2} \int \int \frac{\rho(\mathbf{r}) \rho(\mathbf{r}') g_\lambda(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$

$$\langle \tilde{V}_{ee} \rangle = \frac{1}{2} \int \int \frac{\rho(\mathbf{r}) \rho(\mathbf{r}') \tilde{g}(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad ; \quad \tilde{g}(\mathbf{r}, \mathbf{r}') = \int_0^1 d\lambda g_\lambda(\mathbf{r}, \mathbf{r}')$$

2. Density functional theory

Kohn-Sham density functional theory

$$\tilde{E}_{XC}[\rho] = \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} [\tilde{g}(\mathbf{r}, \mathbf{r}') - 1]$$

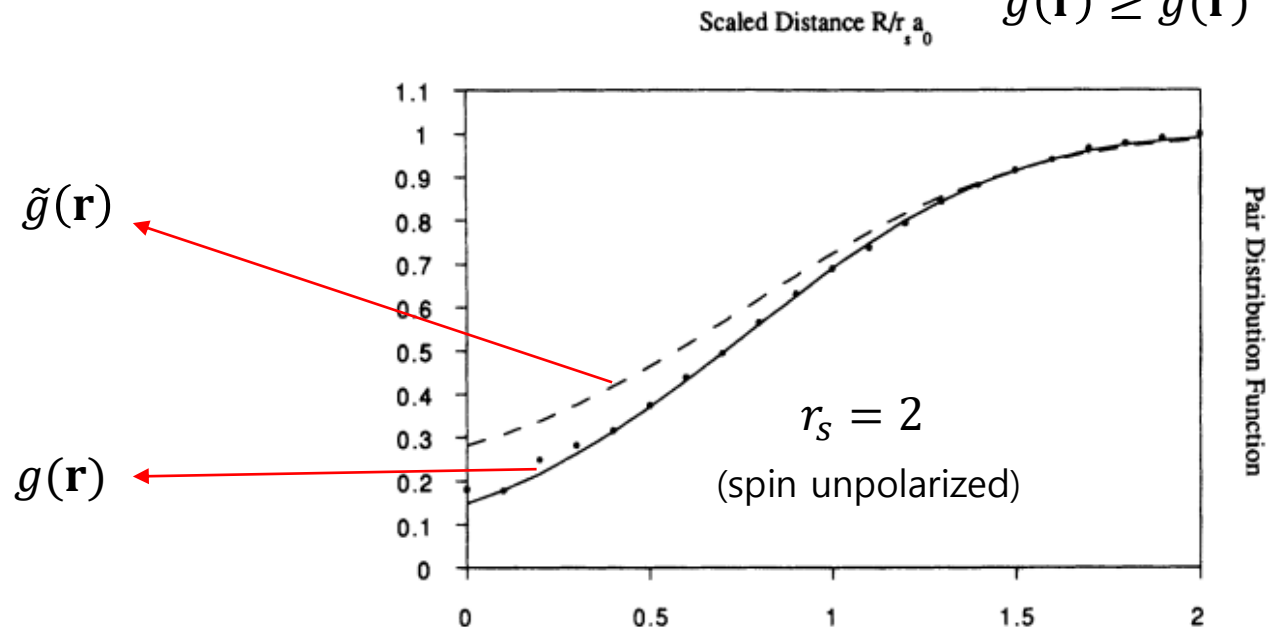
$$\tilde{E}_{XC}[\rho] = E_{XC}[\rho] + \underbrace{T[\rho] - T_{eff}[\rho]}_{\tilde{g}(\mathbf{r}) \geq g(\mathbf{r})}$$

$$\tilde{g}(\mathbf{r}, \mathbf{r}') = g_X(\mathbf{r}, \mathbf{r}') + \tilde{g}_C(\mathbf{r}, \mathbf{r}')$$

$$g_X(\mathbf{r}, \mathbf{r}') = 1 - \frac{\sum_{\sigma} |\rho_{\sigma}(\mathbf{r}, \mathbf{r}')|^2}{\rho(\mathbf{r})\rho(\mathbf{r}')}$$

$$\rightarrow 1 - \frac{9}{2} \left[\frac{j_1(k_F |\mathbf{r} - \mathbf{r}'|)}{k_F |\mathbf{r} - \mathbf{r}'|} \right]^2$$

(homogeneous electron gas)



Perdew and Wang, Phys. Rev. B (1992)

2. Density functional theory)

Kohn-Sham density functional theory

$$\tilde{\rho}_{xc}(\mathbf{r}, \mathbf{r}') = \rho(\mathbf{r}')[\tilde{g}(\mathbf{r}, \mathbf{r}') - 1] \quad \text{"exchange-correlation hole"}$$

- a nonlocal charge depletion due to exchange and correlation, also taking into account kinetic correlation

$$\tilde{E}_{xc}[\rho] = \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r}) \tilde{\rho}_{xc}(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

- $\tilde{g}(\mathbf{r}, \mathbf{r}') = \tilde{g}(\mathbf{r}', \mathbf{r})$
- $\int d\mathbf{r}' \tilde{g}(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') = \int d\mathbf{r} \tilde{g}(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}) = N - 1$
- $\int d\mathbf{r}' \tilde{\rho}_{xc}(\mathbf{r}, \mathbf{r}') = \int d\mathbf{r} \tilde{\rho}_{xc}(\mathbf{r}, \mathbf{r}') = -1$

2. Density functional theory

Kohn-Sham density functional theory

from exchange hole to exchange-correlation hole (Overhauser model) :

$$\left[\frac{d^2}{dr^2} - \frac{l(l+1)}{r^2} + k^2 \right] u_l(r) = 0$$

$u_l(r) = krj_l(kr) \rightarrow$ exchange hole in homogeneous electron gas

$$V(r, r_s) = \begin{cases} \frac{1}{r_s} \left(\frac{r_s}{r} + \frac{r^2}{2r_s^2} - \frac{3}{2} \right), & r \leq r_s \\ 0 & r \geq r_s \end{cases}$$
$$; g_X(\mathbf{r}, \mathbf{r}') = 1 - \frac{9}{2} \left[\frac{j_1(k_F |\mathbf{r} - \mathbf{r}'|)}{k_F |\mathbf{r} - \mathbf{r}'|} \right]^2$$

$$\left[\frac{d^2}{dr^2} - \frac{l(l+1)}{r^2} + k^2 - V(r, r_s) \right] u_l(r) = 0$$

$\rightarrow$ exchange-correlation hole in homogeneous electron gas

2. Density functional theory

Kohn-Sham density functional theory

Thomas-Fermi theory vs. Kohn-Sham theory :

Thomas-Fermi theory

- orbital-free DFT
- kinetic energy as a local term in the density
- limitedly applicable to smooth density, as in alkali metals
- inverting a single equation

⋮

Kohn-Sham theory

- orbital-based DFT
- nonlocal character of kinetic term considered
- not limited
- solving an eigenvalue problem for N lowest eigenvalues

⋮

2. Density functional theory

Kohn-Sham density functional theory

Hartree or Hartree-Fock equation vs. Kohn-Sham equation :

Hartree equation	$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) + V_H \right] \phi_i(\mathbf{r}) = \epsilon_i \phi_i(\mathbf{r})$ • Schrödinger equation for true electron wave function
Hartree-Fock equation	$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) + V_H + K \right] \phi_i(\mathbf{r}) = \epsilon_i \phi_i(\mathbf{r})$ • Schrödinger equation for true electron wave function
Kohn-Sham equation	$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{eff}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \epsilon_i \phi_i(\mathbf{r})$ • Not true electron wave function

2. Density functional theory

Kohn-Sham density functional theory

More about Kohn-Sham theory :

- The main difference between Kohn-Sham and Hartree equations is that $V_{eff}(\mathbf{r})$ includes exchange and correlation.
- In a view of the computational cost ;

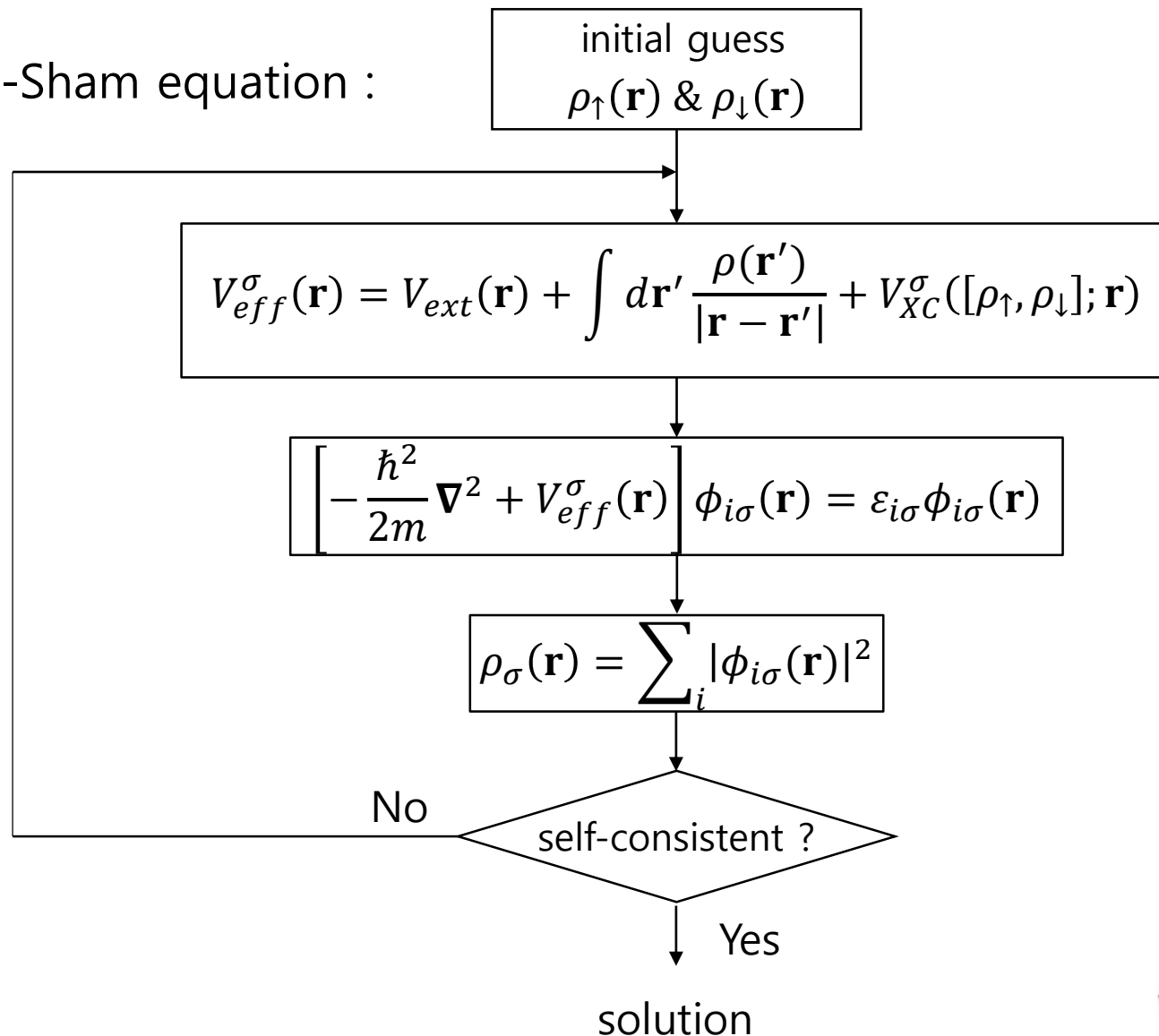
$$[\text{Hartree-Fock}] \gg [\text{Kohn-Sham}] \approx [\text{Hartree}]$$

- The true many-body wave function of the interacting system is not the Slater determinant of Kohn-Sham orbitals. However, it is determined by the density.
- Eigenvalues are not the true energies for adding or subtracting electrons. But the highest eigenvalue (HOMO energy or ionization energy) in a finite system must be correct.

2. Density functional theory

Kohn-Sham density functional theory

Self-consistent loop of Kohn-Sham equation :



0. Structure of matter
1. Many-body theory of electron system
2. Density functional theory
3. Exchange and correlation in DFT
4. Practical solving of electronic problem

3. Exchange and correlation in DFT

Homogeneous electron gas

$$E[\rho] = T_{eff} + V_{ext} + E_H + E_X + \tilde{E}_C$$

$$E_\alpha[\rho] = \int d\mathbf{r} \rho(\mathbf{r}) \epsilon_\alpha[\rho]$$

energy density of contribution α

exchange :

$$\epsilon_X[\rho] = -\frac{3}{4} \left(\frac{3}{\pi} \right)^{1/3} \rho^{1/3} = -\frac{0.458}{r_s}$$

correlation :

$$\epsilon_C[\rho] = \begin{cases} A \ln r_s + B + C r_s \ln r_s + D r_s, & r_s \leq 1 \quad (\text{high density}) \\ \gamma / (1 + \beta_1 \sqrt{r_s} + \beta_2 r_s), & r_s \geq 1 \quad (\text{low density}) \end{cases}$$

Gell-Mann - Brueckner

	A	B	C	D	γ	β_1	β_2
unpolarized gas	0.0311	-0.048	0.002	-0.0116	-0.1423	1.0529	0.3334
fully polarized gas	0.01555	-0.0269	0.0007	-0.0048	-0.0843	1.3981	0.2611

S. Misawa,
Phys. Rev. (1965)

D. Ceperley and B. Alder,
Phys. Rev. Lett. (1980)

J. Perdew and A. Zunger
Phys. Rev. B. (1981)

3. Exchange and correlation in DFT

Homogeneous electron gas

- It may not seem a good idea to use the homogeneous electron gas as a reference for nonuniform systems, for instance, molecular systems.
- Another key issue is that the really meaningful quantity is the sum of the two terms

$$E_X + \tilde{E}_C ,$$

treating both terms to a similar level of approximation ("compatibility problem").

3. Exchange and correlation in DFT

Local density approximation (LDA)

- to use the exchange-correlation hole corresponding to the homogeneous electron gas

$$E_{XC}^{LDA}[\rho] = \int d\mathbf{r} \rho(\mathbf{r}) \epsilon_{XC}^{LDA}[\rho] \quad ; \quad \underbrace{\epsilon_{XC}^{LDA}[\rho] = \epsilon_X[\rho] + \epsilon_C[\rho]}$$

adopted from results of
homogeneous electron gas

$$\epsilon_{XC}^{LDA}[\rho] = \frac{1}{2} \int d\mathbf{r}' \frac{\rho_{XC}^{LDA}(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \quad ; \quad \rho_{XC}^{LDA}(\mathbf{r}, \mathbf{r}') : \text{LDA exchange-correlation hole}$$

$$V_{XC}^{LDA}(\mathbf{r}) = \frac{\delta E_{XC}^{LDA}[\rho]}{\delta \rho} = \epsilon_{XC}^{LDA}[\rho] + \rho(\mathbf{r}) \frac{\delta \epsilon_{XC}^{LDA}[\rho]}{\delta \rho}$$

3. Exchange and correlation in DFT

Local density approximation (LDA)

Local spin density approximation (LSDA) :

$$\rho(\mathbf{r}) = \rho_{\uparrow}(\mathbf{r}) + \rho_{\downarrow}(\mathbf{r}), \quad \zeta(\mathbf{r}) = \rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r})$$

$$\begin{aligned} E_{XC}^{LSDA}[\rho_{\uparrow}(\mathbf{r}), \rho_{\downarrow}(\mathbf{r})] &= \int dr \rho(\mathbf{r}) \epsilon_{XC}^{LSDA}[\rho_{\uparrow}(\mathbf{r}), \rho_{\downarrow}(\mathbf{r})] \\ &= \int dr \rho(\mathbf{r}) \epsilon_{XC}^{LSDA}[\rho(\mathbf{r}), \zeta(\mathbf{r})] \end{aligned}$$

$$\begin{aligned} V_{XC}^{LSDA, \sigma}([\rho_{\uparrow}, \rho_{\downarrow}]; \mathbf{r}) &= \frac{\delta E_{XC}^{LSDA}[\rho_{\uparrow}, \rho_{\downarrow}]}{\delta \rho_{\sigma}(\mathbf{r})} \\ &= \epsilon_{XC}^{LSDA}[\rho_{\uparrow}(\mathbf{r}), \rho_{\downarrow}(\mathbf{r})] + \rho(\mathbf{r}) \frac{\delta \epsilon_{XC}^{LSDA}[\rho_{\uparrow}, \rho_{\downarrow}]}{\delta \rho_{\sigma}(\mathbf{r})} \end{aligned}$$

3. Exchange and correlation in DFT

Local density approximation (LDA)

Local spin density approximation (LSDA) :

$$\epsilon_{XC}^{LSDA}[\rho(\mathbf{r}), \zeta(\mathbf{r})] = \underbrace{f(\zeta)\epsilon_{XC}^U[\rho]} + [1 - f(\zeta)]\underbrace{\epsilon_{XC}^P[\rho]}$$

adopted from unpolarized
homogeneous electron gas

adopted from fully polarized
homogeneous electron gas

$$f(\zeta) = \frac{(1 + \zeta)^{4/3} + (1 - \zeta)^{4/3} - 2}{2^{4/3} - 2}$$

; U. von Barth and L. Hedin, J. Phys. C (1972)

3. Exchange and correlation in DFT

Local density approximation (LDA)

performance of LDA and LSDA:

- LSDA results of H atom

in exact functional,

$$E_H + E_X = 0 \quad (E_H = 0.3125, E_X = -0.3125), \quad E_C = 0$$

in LSDA,

$$E_H + E_X \neq 0 \quad (E_H = 0.3125, E_X = -0.269), \quad E_C \approx -0.03$$

$$\Delta E_X = E_X^{exact} - E_X^{LDA} = -0.0435, \quad \Delta E_C = E_C^{exact} - E_C^{LDA} = 0.03$$

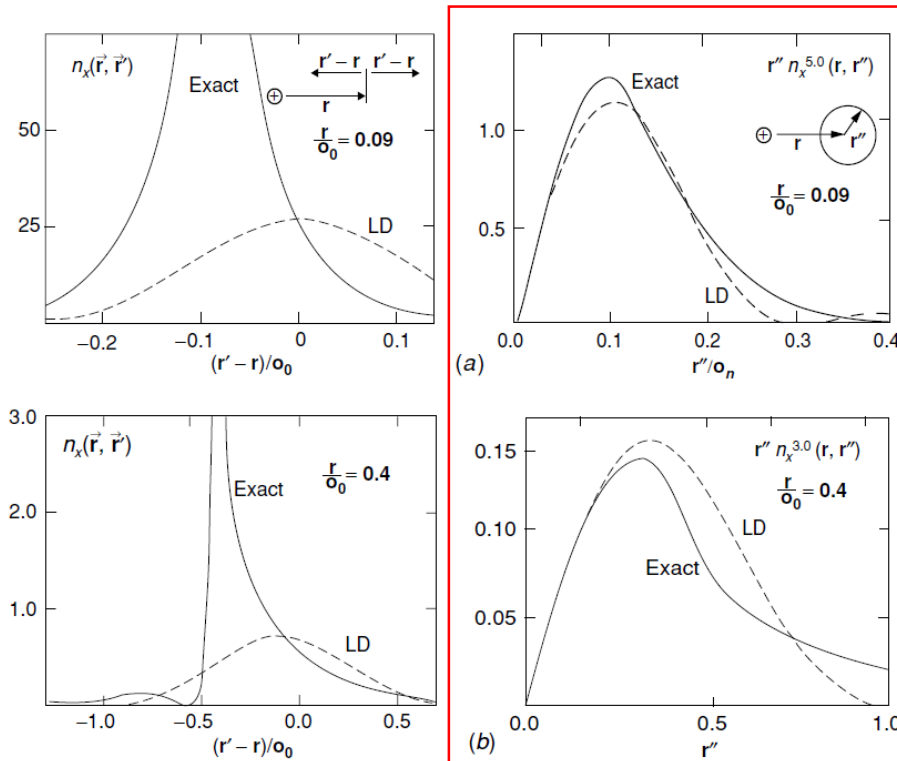
$$\Delta E_{XC} = \Delta E_X - \Delta E_C = -0.0135$$

; errors are partially cancelled in E_{XC}

3. Exchange and correlation in DFT

Local density approximation (LDA)

performance of LDA and LSDA:



$$E_{XC}[\rho] = -\frac{1}{2} \int d\mathbf{r} \rho(\mathbf{r}) \int d\mathbf{r}' \frac{\rho_{XC}(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

spherical average ;

$$\rho_{XC}(\mathbf{r}, \mathbf{r}') \rightarrow \frac{1}{4\pi} \int d\Omega' \rho_{XC}(\mathbf{r}, \mathbf{r}') = \rho_{XC}^{SA}(\mathbf{r}, |\mathbf{r} - \mathbf{r}'|)$$

$$E_{XC}[\rho] = -\frac{1}{2} \int d\mathbf{r} \rho(\mathbf{r}) \times \underbrace{4\pi \int ds s \rho_{XC}^{SA}(\mathbf{r}, s)}$$

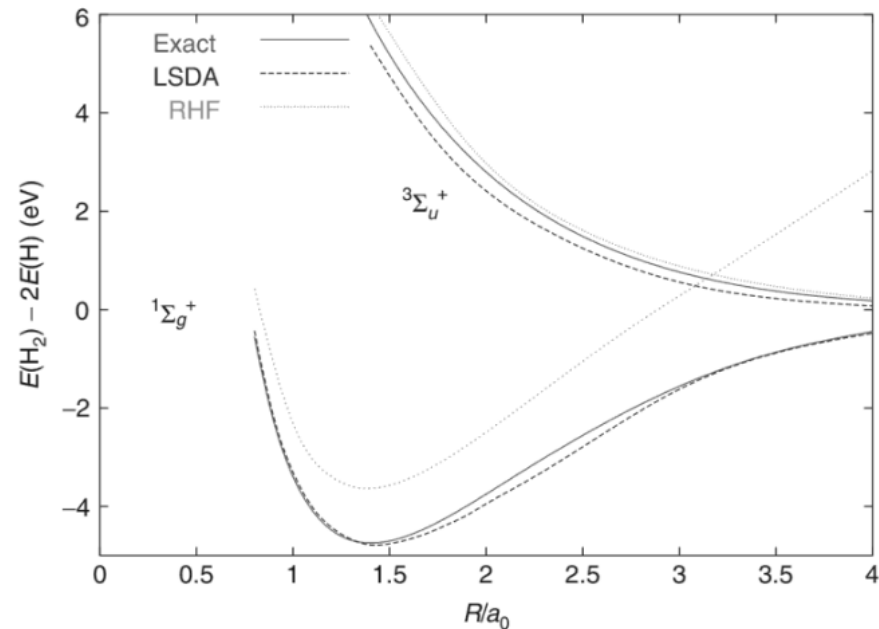
Ne atom, O. Gunnarsson et al., Phys. Rev. B (1979)

- LSDA exchange-correlation energies could be quite reasonable even in atoms
- LSDA exchange-correlation potential is not that good.

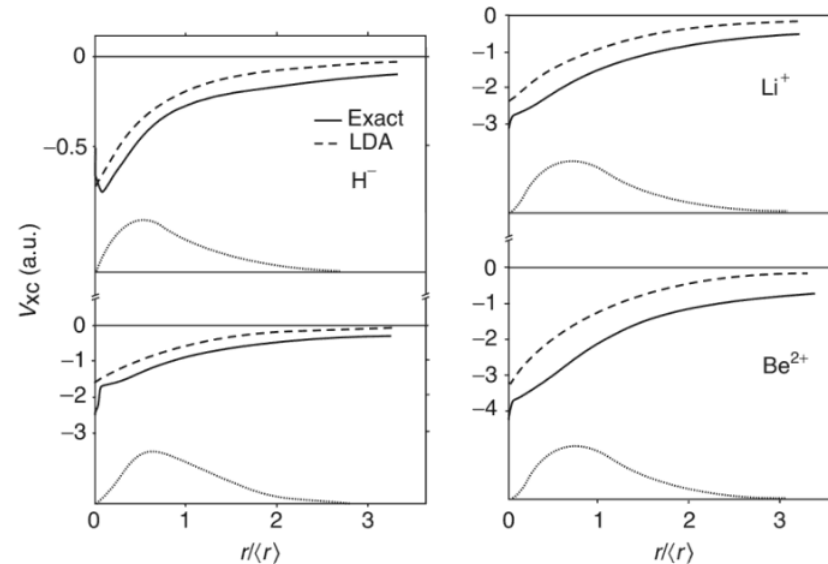
3. Exchange and correlation in DFT

Local density approximation (LDA)

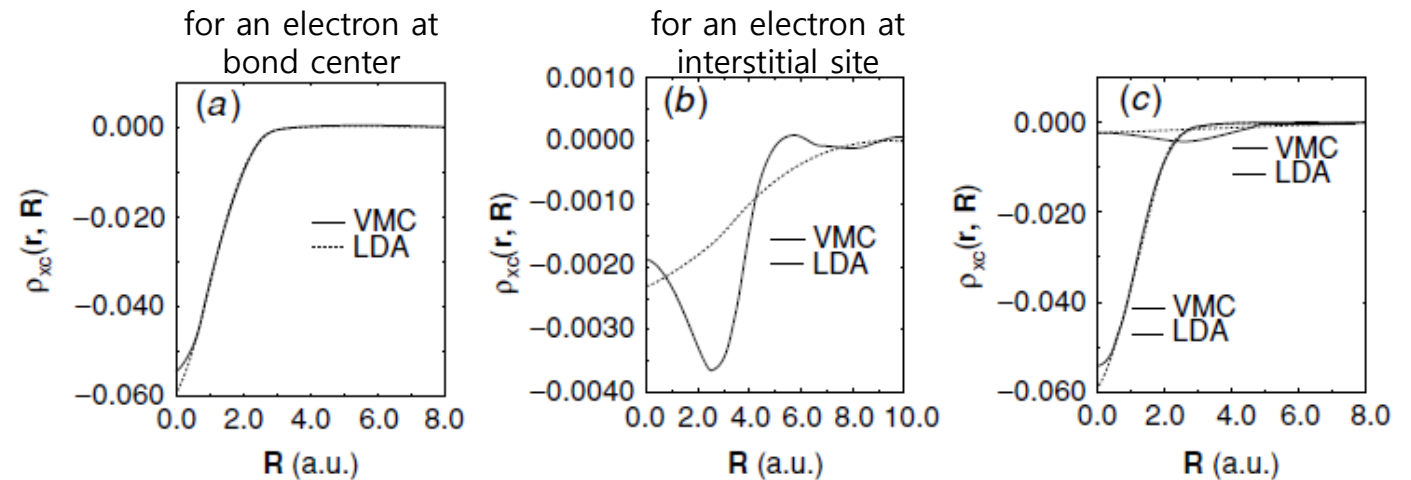
performance of LDA and LSDA:



H_2 molecule,
courtesy by O. Gunnarsson



C. Almbladh and A. Pedroza, Phys. Rev. A (1984)



Si solids, R. Hood et al., Phys. Rev. B (1998)

3. Exchange and correlation in DFT

Local density approximation (LDA)

general features of LDA and LSDA:

- LDA favors the electronic densities that are more homogeneous than exact ones, i.e., LDA underestimates the spatial inhomogeneity of the density than it really is.
- LDA tends to overestimate the binding energy of molecules or the cohesive energy of solids.
 - LDA tends to underestimate the bonding lengths, elastic constants, phonon frequencies.
- LSDA is usually superior to LDA for atomic calculations.

3. Exchange and correlation in DFT

Local density approximation (LDA)

limitations of LDA and LSDA:

- Inhomogeneities in the density are not considered.
- LDA fails to cancel the self-interaction. Hartree-Fock, where the self-interaction is exactly cancelled by the exchange term, gives much better results for atoms. Self-interaction correction (SIC) functional to improve LDA is sometimes needed.

- Energy bandgap E_g in semiconductors is seriously underestimated in LDA

$$E_g = \varepsilon_g + \Delta,$$

where $\varepsilon_g (= \varepsilon_{N+1} - \varepsilon_N)$ is the Kohn-Sham eigenvalue bandgap.

→ Discontinuity in exchange-correlation potential,
i.e., $V_{XC}(N+1) - V_{XC}(N)$ is not properly accounted.

Energy bandgaps (eV)

	LDA	GW	Exp.
diamond	3.9	5.6	5.48
Si	0.52	1.29	1.17
Ge	0.07	0.75	0.74
LiCl	6.0	9.1	9.4

M. Hybertsen and S. Louie,
Phys. Rev. B (1986)

3. Exchange and correlation in DFT

Gradient expansion

$$E_{XC}[\rho] = \int d\mathbf{r} \rho(\mathbf{r}) \epsilon_{XC}[\rho(\mathbf{r})] F_X[\rho(\mathbf{r}), \nabla\rho(\mathbf{r}), \nabla^2\rho(\mathbf{r}), \dots]$$

exchange enhancement factor :

$$F_X(s_1, s_2) = 1 + \frac{10}{81} s_1^2 + \frac{146}{2025} s_2^2 + \dots$$

$$s_1 = \frac{|\nabla\rho|}{2(3\pi^2)^{1/3} \rho^{4/3}}$$

$$s_2 = \frac{|\nabla^2\rho|}{4(3\pi^2)^{2/3} \rho^{5/3}}$$

$$F_X(s_1) \approx 1 + \frac{10}{81} s_1^2$$

→ generalized gradient approximation (GGA)

3. Exchange and correlation in DFT

Generalized gradient approximation (GGA)

$$E_{XC}^{GGA}[\rho] = \int d\mathbf{r} \rho(\mathbf{r}) \epsilon_{XC}^{GGA}[\rho] = \int d\mathbf{r} A_{XC}[\rho] \rho(\mathbf{r})^{4/3} + \int d\mathbf{r} C_{XC}[\rho] |\nabla \rho(\mathbf{r})|^2 / \rho(\mathbf{r})^{4/3}$$

$$E_{XC}^{GGA}[\rho_{\uparrow}, \rho_{\downarrow}, \nabla \rho_{\uparrow}, \nabla \rho_{\downarrow}] = \int d\mathbf{r} \rho(\mathbf{r}) \epsilon_{XC}^{GGA}[\rho_{\uparrow}, \rho_{\downarrow}, \nabla \rho_{\uparrow}, \nabla \rho_{\downarrow}] = \frac{1}{2} (E_{XC}^{GGA}[\rho_{\uparrow}, \nabla \rho_{\uparrow}] + E_{XC}^{GGA}[\rho_{\downarrow}, \nabla \rho_{\downarrow}])$$

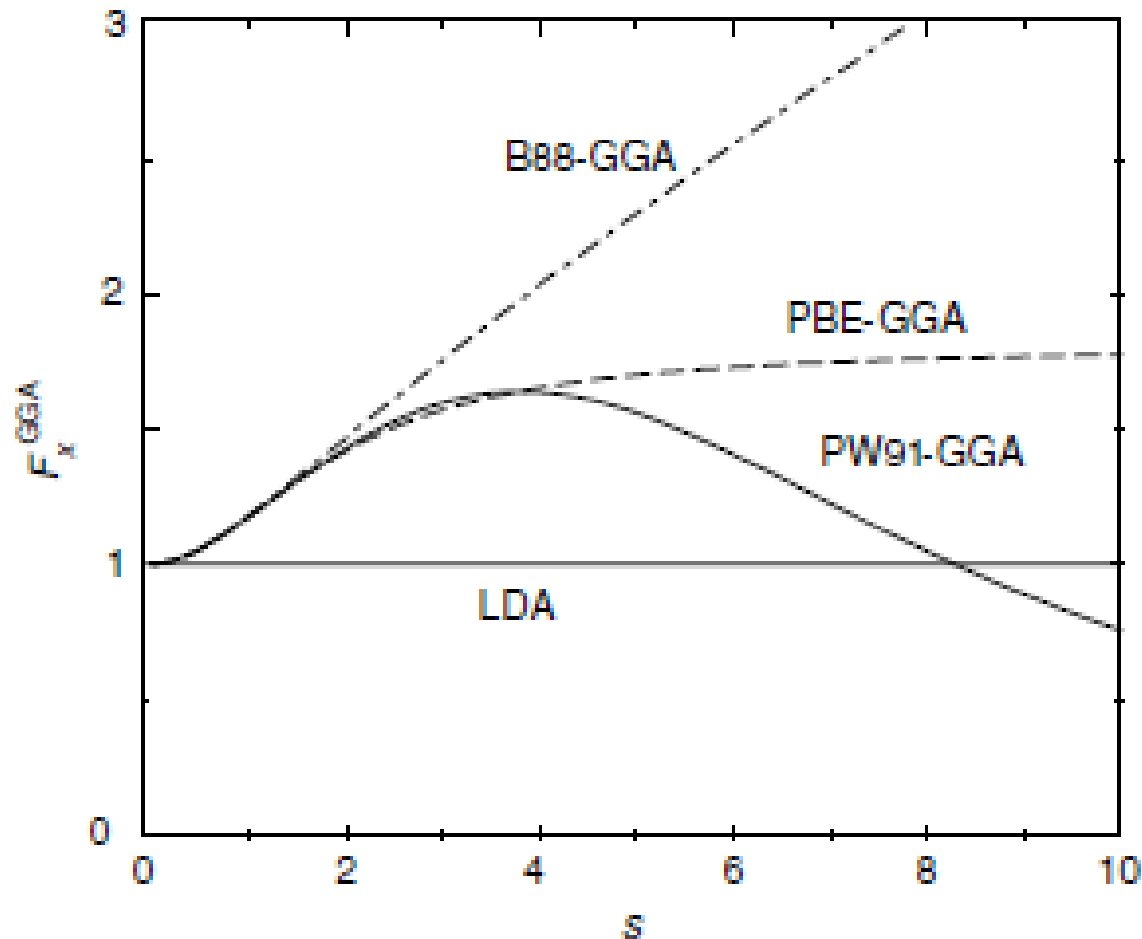
$$\begin{aligned} V_{XC}^{GGA, \sigma}([\rho_{\uparrow}, \rho_{\downarrow}, \nabla \rho_{\uparrow}, \nabla \rho_{\downarrow}]; \mathbf{r}) &= \frac{\delta E_{XC}^{GGA}[\rho_{\uparrow}, \rho_{\downarrow}, \nabla \rho_{\uparrow}, \nabla \rho_{\downarrow}]}{\delta \rho_{\sigma}(\mathbf{r})} \\ &= \epsilon_{XC}^{GGA}[\rho_{\uparrow}, \rho_{\downarrow}, \nabla \rho_{\uparrow}, \nabla \rho_{\downarrow}] + \rho(\mathbf{r}) \frac{\delta \epsilon_{XC}^{GGA}[\rho_{\uparrow}, \rho_{\downarrow}, \nabla \rho_{\uparrow}, \nabla \rho_{\downarrow}]}{\delta \rho_{\sigma}(\mathbf{r})} \\ &\quad - \nabla \left(\rho(\mathbf{r}) \frac{\delta \epsilon_{XC}^{GGA}[\rho_{\uparrow}, \rho_{\downarrow}, \nabla \rho_{\uparrow}, \nabla \rho_{\downarrow}]}{\delta \nabla \rho_{\sigma}(\mathbf{r})} \right) \end{aligned}$$

→ a straightforward evaluation is ill-behaved, mathematically difficult

3. Exchange and correlation in DFT

Generalized gradient approximation (GGA)

- a number of modified expansions are proposed



Becke : A. Becke, Phys. Rev. A (1988)

PW91 : J. Perdew and Y. Wang, Phys. Rev. B (1988)

PBE : J. Perdew., K. Burke, and M. Ernzerhof, Phys. Rev. Lett. (1996)

3. Exchange and correlation in DFT

Generalized gradient approximation (GGA)

Langreth-Mehl functional :

- the first GGA

$$\epsilon_X = \epsilon_X^{LDA} - a \frac{|\nabla\rho(\mathbf{r})|^2}{\rho(\mathbf{r})^{7/3}} \left(\frac{7}{9} + 18f^2 \right)$$

$$\epsilon_C = \epsilon_C^{RPA} - a \frac{|\nabla\rho(\mathbf{r})|^2}{\rho(\mathbf{r})^{7/3}} (2e^{-F} + 18f^2)$$

$$; F = b |\nabla\rho(\mathbf{r})|/\rho(\mathbf{r})^{7/6}$$

$$b = (2\pi)^{1/6} f, \quad f = 0.15$$

$$a = \pi/16(3\pi^2)^{4/3}$$

3. Exchange and correlation in DFT

Generalized gradient approximation (GGA)

Becke exchange functional :

$$\epsilon_X = \epsilon_X^{LDA} \left(1 - \frac{\beta}{2^{1/3} A_x} \frac{x^2}{1 + 6\beta x \sinh^{-1} x} \right) \quad ; \quad x = 2^{1/3} |\nabla \rho(\mathbf{r})| / \rho(\mathbf{r})^{4/3}, \quad A_x = (3/4)(3/\pi)^{1/3} \\ \beta = 0.0042$$

Lee, Yang, and Parr (LYP) correlation functional :

- not LDA based

$$\epsilon_C = -\frac{a}{1 + d\rho^{-1/3}} \left\{ \rho + b\rho^{-2/3} \left[C_F \rho^{5/3} - 2t_W + \frac{1}{9} \left(t_W + \frac{1}{2} \nabla^2 \rho \right) \right] e^{-c\rho^{-1/3}} \right\} \\ ; \quad t_W = \frac{1}{8} \left(\frac{|\nabla \rho|^2}{\rho} - \nabla^2 \rho \right), \quad C_F = 3/10 (3\pi^2)^{2/3}, \quad a = 0.04918, \quad b = 0.132, \quad c = 0.2533 \\ d = 0.349$$

 BLYP functional
combination

3. Exchange and correlation in DFT

Generalized gradient approximation (GGA)

Perdew, Burke, and Ernzerhof (PBE) exchange functional :

$$F_X(s) = 1 + \kappa - \frac{\kappa}{1 + \mu s^2 / \kappa} \quad ; \quad s = |\nabla \rho| / (2k_F \rho), \quad \mu = \beta(\pi^2/3) = 0.21951 \\ \beta = 0.066725$$

$F_X(s)$ satisfies

- recovers the correct homogeneous electron gas limit, i.e., $F_X(0) = 1$
- recovers the LSDA linear response limit for $s \rightarrow 0$, i.e., $F_X(s) \rightarrow 1 + \mu s^2$
- the local Lieb-Oxford inequality, $\epsilon_{XC}(\mathbf{r}) \geq -1.679\rho(\mathbf{r})^{1/3}$, provided that $\kappa \leq 0.804$

$$; \text{ Lieb-Oxford inequality (stability of matter)} \rightarrow \underline{E_{XC}} \geq -1.679 \int d\mathbf{r} \rho(\mathbf{r})^{4/3}$$

total Coulomb energy — classical Coulomb energy

3. Exchange and correlation in DFT

Generalized gradient approximation (GGA)

$$\epsilon_C^{PBE}[\rho] = \epsilon_C^{LDA}[\rho, \zeta] + H[\rho, \zeta, t]$$

$$H[\rho, \zeta, t] = \gamma \phi^3 \ln \left[1 + \frac{\beta}{\gamma} t^2 \frac{1 + At^2}{1 + At^2 + A^2 t^4} \right] \quad ; \quad t = |\nabla \rho| / (2\phi k_s \rho) , \quad k_s \rightarrow \text{Thomas - Fermi wave vector}$$
$$\phi(\zeta) = [(1 + \zeta)^{2/3} + (1 - \zeta)^{2/3}] / 2$$
$$\beta = 0.066725 , \quad \gamma = (1 - \ln 2) / \pi^2$$
$$A = \frac{\beta}{\gamma} \left[e^{-\epsilon_C^{LDA}[\rho] / \gamma \phi^3} - 1 \right]^{-1}$$

$H[\rho, \zeta, t]$ satisfies

- recovers the correct second-order gradient expansion in the slowly varying limit, i.e., $t \rightarrow 0$
- approaches minus the homogeneous electron gas correlation $-\epsilon_C^{LDA}$ for rapidly varying densities ($t \rightarrow \infty$), i.e., $\epsilon_C^{PBE} \rightarrow 0$
- cancels the logarithmic singularity in ϵ_C^{LDA}

3. Exchange and correlation in DFT

Generalized gradient approximation (GGA)

General practices of GGA :

- GGA improves binding energies and bond lengths for molecules.
- For 4d-5d transition metals (including semiconductors), the improvement of GGA over LDA is not clear.
- There is some improvement in the bandgap, but not substantial.
 - Discontinuity Δ in exchange-correlation potential still vanishes, but the Kohn-Sham eigenvalue bandgap increases compared to LDA.
- GGA does not satisfy asymptotic behaviors for isolated atoms.
 - $V_{XC}(\mathbf{r}) \rightarrow -e^2/r$ as $r \rightarrow \infty$, while $V_{XC}^{GGA,LDA}(\mathbf{r})$ vanish exponentially.

3. Exchange and correlation in DFT

LDA (GGA) + U

Hubbard phenomenological approach :

- transition metal oxides (d electrons), rare-earth metal oxides (f electrons), ...

$$E_{LDA+U} = E_{LDA} + E_U - \underbrace{\frac{1}{2}UN(N-1)}$$

- removing double counting in E_{LDA}

$$E_U = \frac{U}{2} \sum_i (n_i - n_i^2) \quad ; \text{ S.L. Dudarev et al., Phys. Rev. B (1998)}$$

$$\varepsilon_i = \varepsilon_{LDA} + U \left(\frac{1}{2} - n_i \right)$$

- $n_i = 1$ (occupied) $\rightarrow -U/2$
- $n_i = 0$ (unoccupied) $\rightarrow +U/2$

$$V_i = \frac{\partial E_U}{\partial n_i} = U \left(\frac{1}{2} - n_i \right)$$

3. Exchange and correlation in DFT

GW approximation

Green's function approach :

- electronic excitations available

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(\mathbf{r}) - E \right] G(\mathbf{r}, \mathbf{r}'; E) + \int d\mathbf{r}'' \underbrace{\Sigma(\mathbf{r}, \mathbf{r}''; E)} G(\mathbf{r}'', \mathbf{r}'; E) = -\delta(\mathbf{r} - \mathbf{r}')$$

- self-energy operator,
nonlocal and dynamical exchange and correlation

$$G_0(\mathbf{r}, \mathbf{r}'; E) = \sum_n \frac{\phi_n(\mathbf{r}) \phi_n^*(\mathbf{r}')}{E - \varepsilon_n}$$

- $\phi_n(\mathbf{r})$, ε_n : LDA orbital and energy

$$\overline{\overline{G}} = \overline{\overline{G_0}} + \overline{\overline{G_0}} \text{ (cloud) } \overline{\overline{G_0}}$$

$$\text{wavy } W = \text{dashed} + \text{dashed} \text{ (loop } P \text{) wavy}$$

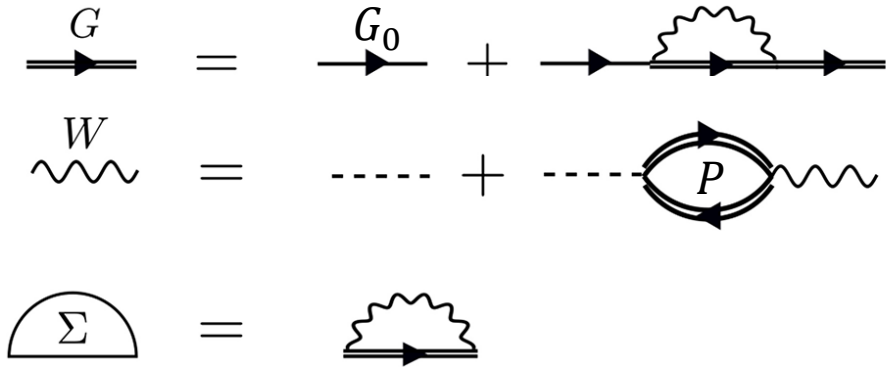
$$\text{semi-circle } \Sigma = \text{cloud}$$

3. Exchange and correlation in DFT

GW approximation

Hedin's way :

L. Hedin, Phys. Rev (1965)



$$\Sigma_{GW}(\mathbf{r}, \mathbf{r}'; E) = i \int dE' G(\mathbf{r}, \mathbf{r}'; E + E') W(\mathbf{r}, \mathbf{r}'; E')$$

$$W(\mathbf{r}, \mathbf{r}'; E) = v(\mathbf{r}, \mathbf{r}') + \int \int d\mathbf{r}_1 d\mathbf{r}_2 W(\mathbf{r}, \mathbf{r}_1; E) P(\mathbf{r}_1, \mathbf{r}_2; E) v(\mathbf{r}_2, \mathbf{r}') \quad ; \quad P(\mathbf{r}, \mathbf{r}'; E) = -i G(\mathbf{r}, \mathbf{r}'; E) G(\mathbf{r}', \mathbf{r}; E)$$

- $\epsilon(\mathbf{r}, \mathbf{r}_1; E)$: dielectric function

→ plasmon pole approximation

(M. Hybertsen and S. Louie, Phys. Rev. Lett. (1985))

→ full RPA

(F. Aryasetiawan and O. Gunnarsson, Phys. Rev. Lett. (1995))

$$W(\mathbf{r}, \mathbf{r}'; E) = \int d\mathbf{r}_1 \frac{v(\mathbf{r}_1, \mathbf{r}')}{\epsilon(\mathbf{r}, \mathbf{r}_1; E)}$$

- Vertex correction is sometimes need for the sum rule.

- $G_0 W_0, GW_0, GW$ self-consistencies are investigated. (B. Holm, Phys. Rev. Lett. (1999))

3. Exchange and correlation in DFT

Ladder in functionals

•
•
•

Meta GGA

$$\epsilon_{XC}[\rho(\mathbf{r}), \nabla\rho(\mathbf{r}), \tau(\mathbf{r})]$$

; $\tau(\mathbf{r})$: kinetic
energy density

Generalized gradient approximation (GGA)

$$\epsilon_{XC}[\rho(\mathbf{r}), \nabla\rho(\mathbf{r})]$$

Local density approximation (LDA)

$$\epsilon_{XC}[\rho(\mathbf{r})]$$

0. Structure of matter
1. Many-body theory of electron system
2. Density functional theory
3. Exchange and correlation in DFT
4. Practical solving of electronic problem

4. Practical solving of electronic problem

Electron-nuclear interaction

- valence electrons – actively participating in chemical bonding
- core electrons – tightly bound to nuclei

valence electrons + core electrons $\rightarrow$ all electron methods

valence electrons + effective nuclear charge $\rightarrow$ pseudopotential methods

4. Practical solving of electronic problem

Basis sets

- extended basis set – useful for solids
- localized basis set – useful for molecules
- mixed basis set – designed for both solids and molecules
- augmented basis set – extended basis set is augmented with atomic-like wave functions around nuclei

4. Practical solving of electronic problem

k-space formulation of Kohn-Sham equation

- under the periodic potential

$$\phi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} \sum_{\mathbf{G}} c_{\mathbf{k}+\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}} \quad ; \text{ plane wave basis}$$

$$H_{\mathbf{G},\mathbf{G}'}(\mathbf{k}) = \frac{\hbar^2}{2m} |\mathbf{k} + \mathbf{G}|^2 \delta_{\mathbf{G},\mathbf{G}'} + V_{eff}(\mathbf{G} - \mathbf{G}')$$

$$\sum_{\mathbf{G}'} H_{\mathbf{G},\mathbf{G}'}(\mathbf{k}) c(\mathbf{k} + \mathbf{G}') = \varepsilon(\mathbf{k}) c(\mathbf{k} + \mathbf{G})$$

to be better for core electrons



orthogonalized plane wave (OPW), pseudopotential method,
augmented plane wave (APW) method ...

4. Practical solving of electronic problem

4. Practical solving of electronic problem